A Branch-and-Cut-and-Price Algorithm for the Mixed Capacitated General Routing Problem

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In this paper, we consider the Mixed Capacitated General Routing Problem which is a combination of the Capacitated Vehicle Routing Problem and the Capacitated Arc Routing Problem. The problem is also known as the Node, Edge, and Arc Routing Problem. We propose a Branch-and-Cut-and-Price algorithm for obtaining optimal solutions to the problem and present computational results based on a set of standard benchmark instances.
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1. INTRODUCTION

The Capacitated Vehicle Routing Problem (CVRP) was first presented in [18] and has been studied extensively ever since. The Capacitated Arc Routing Problem (CARP) was presented more than twenty years later in [25]. Both problems find many applications in real life. The CVRP models problems of delivery of products and other situations where specific points have to be visited. The CARP, on the other hand, models situations where street segments need to be visited. Such problems are often found in snow removal and refuse collection, for instance.

In this paper, we study a combination of the two types of problems. In the Mixed Capacitated General Routing Problem (MCGRP), nodes as well as directed arcs and undirected edges require visits from a fleet of vehicles. The MCGRP was first introduced in [39] and was rediscovered in [40] under a different name: The Node, Edge, and Arc Routing Problem (NEARP).

The MCGRP more specifically models situations where street segments as well as specific points need service. Applications are found within refuse collection where household waste is collected along the streets whereas industrial sites and supermarkets need to be considered as single locations due to the large amount of waste in these locations. And due to sheer size, large arc routing problems in real life waste management, snow removal, or mail delivery cannot be handled efficiently. One option for solving such problems is to contract neighborhoods of 10–20 arcs and edges into a single node, while leaving larger streets and highways as arcs and edges. The result of this process is a MCGRP.

Until now only two exact algorithms for the MCGRP have been presented, both being Branch-and-Cut algorithms [15, 30]. Inspired by [23] we present the first Branch-and-Cut-and-Price algorithm for the problem. The main idea is to combine two different mathematical models for the problem, where one model involves cut generation and the other involves column generation. The purpose is to present a robust model where cuts can easily be added in a Branch-and-Price setup. A preliminary version of our algorithm, where cuts were only added in the root node, was presented in [3].

In Section 2, we present relevant literature regarding exact optimization of the CVRP and the CARP as well as the limited literature regarding the MCGRP. In Section 3, we first define the problem formally and introduce some notation. Next we present the two models of which our model is composed, after which we present our combined model. Finally, we describe some valid inequalities which can be used for strengthening the formulation. In Section 4, we present the details of

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our Branch-and-Cut-and-Price algorithm and in Section 5 we provide our computational results. Finally, conclusions are drawn in Section 6.

2. RELATED LITERATURE

As the MCGRP is a combination of node routing and edge/arc routing, we first consider the literature presenting exact solutions of the CVRP and the CARP, respectively. We then turn our attention to previous work related to the MCGRP. The aim is not to provide a full review of papers published within the area, but rather to describe the state of the art.

Many papers have presented exact algorithms for the CVRP. A major breakthrough was the robust Branch-and-Cut-and-Price algorithm in [23]. This algorithm simultaneously generates columns based on a set partitioning formulation of the problem and cuts based on a flow formulation. The cuts are based on valid inequalities presented in [34] and [38].

In [4], an exact set partitioning algorithm for the CVRP is presented, where the LP relaxation of the master problem is solved by sequentially solving three problems. Each of these problems is obtained by tightening the former with a new class of valid inequalities or by requiring routes to be elementary. In [6], this algorithm is improved by a new route generation method and the algorithm is extended to solve the vehicle routing problem with time windows. In [5], the exact algorithm in [4] is extended to solve a broader class of vehicle routing problems, including a heterogeneous fleet of vehicles, multiple depots, and site dependence.

Recent surveys on the CVRP and on exact approaches in particular are provided in [33, 41] and [7]. In [7], the exact methods of [4, 6, 23, 38] are compared. Their results show that, on average, the algorithm in [6] outperforms the algorithms in [4] and [23]. The algorithm in [38] is outperformed by the others but is used as part of the algorithm in [23].

In [9], the authors suggest an IP model for the CARP based on 0-1 variables x_{ep}, representing whether or not vehicle p services edge e, and variables y_{ep} representing the number of times vehicle p traverses edge e without servicing it. They study the polyhedron of the CARP based on this model and present several facets for this polyhedron. In [10], the work is further extended to an aggregated model using variables x_e to represent the number of times the edge e is traversed without service by any vehicle. They use this model in a cutting plane algorithm for obtaining lower bounds. Due to the aggregation, the algorithm cannot be used for obtaining feasible CARP solutions, but it produces high quality lower bounds.

In [37], the CARP is solved by using a transformation to the CVRP and using the Branch-and-Cut-and-Price algorithm in [23] to solve the transformed problem. In [8], the CARP is solved by transforming it to a CVRP and using a series of cut and column generations which is similar to the strategy used in [4].

In [14], the authors present an exact algorithm for the CARP based on a Cut-First Branch-and-Price-Second strategy. Their algorithm solves nearly all unsolved benchmark instances for the problem. In their algorithm, they use the pricing strategy suggested in [35]. They use algorithms from [36] for Odd Edge Cutset, from [10] and [1] for Capacity inequalities, and from [10] for Disjoint-path inequalities. Furthermore, the paper provides a full survey of exact solutions of the CARP and a detailed discussion of different modeling and solution approaches for the problem.

In [12], the authors present a comprehensive analysis of the use of different pricing strategies within the Cut-First Branch-and-Price-Second algorithm from [14]. In [12] methods for handling (k_1, k_2)-loop elimination in a SPPRC are investigated. The k_1 refers to loop elimination with respect to modeling elementary routes, and k_2 refers to eliminating loops which would violate non-follower constraints. In [13], the same authors introduce a SPPRC algorithm with $(k,2)$-loop elimination for efficiently handling the non-follower constraints arising in the Branch-and-Price algorithm presented in [14], this SPPRC algorithm allows for $(k > 3)$-cycle elimination. A recent survey on the CARP and its variations is available in [17].

The MCGRP was first presented in [39]. They motivate the introduction of this problem by real life applications and solve the problem, including both capacity and time duration constraints, using a route-first-partition-second algorithm. In [26], a cluster-first-route-second algorithm is presented for the same problem, and it is shown that their algorithm outperforms that of [39].

Four papers have considered the MCGRP from a metaheuristic point of view: A memetic algorithm [40], an algorithm based on Simulated Annealing [32], an Iterated Local Search algorithm [21], and finally the Spider algorithm [27] is used in [28]. In [16], the first matheuristic is presented for the MCGRP. The algorithm involves a large number of neighborhood structures and draws upon the Branch-and-Cut algorithm presented in [15] for improving substructures of a solution obtained by considering two routes at a time.

To obtain lower bounds, [24] extends the column generation approach in [35] to the MCGRP and uses it to solve the root node of a Branch-and-Price tree. In [2], a lower bounding algorithm is presented for the MCGRP based on the construction of a matching network in subgraphs obtained by a series of mutually disjoint cuts.

Two exact algorithms have been published for the MCGRP. In [15], the authors present a three-index IP model and extend the capacity constraints and the odd edge cutset constraint from [9] to the MCGRP and incorporate them in a Branch-and-Cut algorithm. Moreover, the authors of [15] test five heuristic procedures and use the best of these to initialize their algorithm. Their algorithm can solve small instances to optimality within a time limit of 6 hours. In [30], the authors present a two-phase Branch-and-Cut algorithm, where they first solve an aggregated model to obtain tight lower bounds and strong valid inequalities that are then used to speed up their Branch-and-Cut based on a two-index formulation.

Furthermore, a preliminary version of our algorithm in the present paper was presented in [3].

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3. MODEL FORMULATION

The Mixed Capacitated General Routing Problem (MCGRP) is defined on a connected graph $G = (N, E, A)$, where N is the set of nodes, E is the set of undirected edges, and A is the set of directed arcs. Let c_e (c_a) denote the nonnegative traversal cost for $e \in E$ ($a \in A$). Let $N_R \subseteq N$ be the set of required nodes, i.e., nodes which require service, and let q_i denote the size of the demand for each node $i \in N_R$. Similarly, let E_R and A_R be the set of required edges and arcs, respectively, and let q_e (q_a) denote the demand for each $e \in E_R$ ($a \in A_R$), respectively.

Node $1 \in N$ denotes the depot node and holds a fleet of identical vehicles, each of capacity W. We define $N' = N \backslash \{1\}$. Moreover, we let K denote the set of vehicles. The goal is to construct tours for the vehicles, starting and ending in the depot, such that each required node, arc, and edge is serviced by exactly one vehicle, the vehicle capacity is respected, and the total traversal cost is minimized.

3.1. Notation and Definitions

In our notation, an edge e represents an unordered pair of nodes $\{i, j\}$, and an arc a represents an ordered pair of nodes (i, j). We refer jointly to any arc or edge as a link. For any set $S \subseteq N'$, we define the graph $G(S)$ induced by S as the graph with node set S, edge set $E(S) = \{e = \{i,j\} \in E | i \in S, j \in S\}$, and arc set $A(S) = \{a = (i,j) \in A | i \in S, j \in S\}$.

For any set $S \subset N$, we define the following cutsets: The set of edges crossing the cut $\delta^u(S) = \{e = \{i,j\} \in E | i \in S, j \in N \setminus S \text{ or } i \in N \setminus S, j \in S\}$, the set of outgoing arcs $\delta^+(S) = \{a = (i,j) \in A | i \in S, j \in N \setminus S\}$, and the set of entering arcs $\delta^-(S) = \{a = (i,j) \in A | i \in N \setminus S, j \in S\}$. We define the joint cutset $\delta(S)$ as the total set of links in any of the cutsets:

$$\delta(S) = \delta^u(S) \cup \delta^+(S) \cup \delta^-(S) \subseteq E \cup A$$

Throughout the paper, any set with subscript R denotes the required entities of that set. Given a set $S \subseteq N'$, the total demand in $G(S)$ and $\delta(S)$ is denoted $Q(S)$ and is given by $Q(S) = \sum_{e \in E_R(S) \cup A_R(S) \cup \delta_R(S)} q_e + \sum_{i \in S_R} q_i$. To service this demand, at least $\gamma(S) = \lceil Q(S)/W \rceil$ vehicles are needed. In other words, at least $\gamma(S)$ vehicles must enter the set S. When S consists of a single node i, we simplify notation and write i instead of $\{i\}$ in all the above definitions.

3.2. Set Partitioning Model

Models for the MCGRP based on flow variables are presented in [15, 30]. Using Dantzig-Wolfe decomposition [19, 20] on such a model provides us with a Set Partitioning model for the problem. Because all vehicles in our model are identical, we can simplify the decomposition by defining a common set of routes for all vehicles. Specifically, we let Ω denote the set of routes for a single vehicle with the following requirements for each route: (1) The vehicle must start and end in the depot, (2) The vehicle must service one or more required entities, (3) Each entity is serviced at most once on the route, (4) Any deadheading between two serviced entities, and between the depot and the first (last) entity serviced must follow a shortest path, and (5) The vehicle capacity must be respected.

The third requirement states that the route must be elementary with respect to serviced entities. In practice, this requirement is often relaxed to simplify the column generation process for the pricing problem. In this case, the requirement is handled in a branching process.

It is essential that the deadheading occurs along a shortest path. If this requirement is relaxed, the problem is unbounded from above. An alternative is to leave the feasible region unbounded and include extreme rays consisting of small deadheading loops. See [14] for a detailed description of this.

We let c_r be the total traversal cost of route r and define the following parameters:

$h_i^r = 1$ if route r services node i, and 0 otherwise.

$s_e^r = 1$ if route r services edge e, and 0 otherwise.

$t_a^r = 1$ if route r services arc a, and 0 otherwise.

To construct the Master Problem, we define, for each route $r \in \Omega$, the variable

$$\lambda^r = \begin{cases} 1 & \text{if route } r \text{ is used in the solution} \\ 0 & \text{otherwise} \end{cases}$$

The Master Problem can then be stated as

(IPMaster)

$$\text{minimize} \quad \sum_{r \in \Omega} c_r \lambda^r$$

$$\text{s.t.} \quad \sum_{r \in \Omega} h_i^r \lambda^r = 1, \forall i \in N_R \tag{1}$$

$$\sum_{r \in \Omega} s_e^r \lambda^r = 1, \forall e \in E_R \tag{2}$$

$$\sum_{r \in \Omega} t_a^r \lambda^r = 1, \forall a \in A_R \tag{3}$$

$$\sum_{r \in \Omega} \lambda^r \leq |K| \tag{4}$$

$$\lambda^r \geq 0 \text{ and integer } \forall r \in \Omega \tag{5}$$

In (IP Master), constraints (1), (2), and (3) ensure that every required node, edge, and arc is serviced by exactly one vehicle route, and constraint (4) ensures that no more than $|K|$ vehicles are used. For the LP relaxation of (IP Master), let $\pi = [\pi_i]$, $\mu = [\mu_e]$, and $\rho = [\rho_a]$ be the vectors of dual variables corresponding to constraints (1), (2), and (3), respectively. Finally, let σ be the dual variable corresponding to constraint (4). Then the pricing problem is given by the following (IP Pricing).

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(IP Pricing) Consider a graph with costs $-\pi_i$, $c_e - \mu_e$, and $c_a - \rho_a$ for servicing nodes i, edges e, and arcs a, respectively, and costs c_e and c_a for deadheading edges and arcs, respectively. Determine in this graph the minimum cost elementary path, starting and ending at the depot, subject to the vehicle capacity constraint.

The cost of the path identified in (IP Pricing) differs only by a constant from the reduced cost of the corresponding column in (IP Master), hence (IP Pricing) identifies a column with minimum reduced cost. By subtracting σ from the path cost in (IP Pricing) we obtain the reduced cost in (IP Master).

3.3. Aggregated Model

In [10], a one-index model for a relaxation of the CARP is presented. The model is based on aggregated variables defined as

$$x_e = \sum_{k \in K} x_{ek} \; \forall e \in E$$

where x_{ek} represents the number of times edge e is deadheaded by vehicle k. Intuitively, x_e counts the number of times an edge is deadheaded by any vehicle. The drawback of defining a model based on such aggregated variables is that an optimal solution to the model may not necessarily correspond to a feasible solution to the CARP. The advantage, on the other hand, is that the number of variables is smaller than in the two-index model. A similar type of model is used for the MCGRP in [30].

In [10], several valid inequalities are presented based on these aggregated variables and used in a Cutting Plane algorithm that successfully obtains lower bounds for the CARP.

We use the idea of aggregated variables in our combined model which is presented in Section 3.4. In Section 3.5 we present how valid inequalities in [10] are extended to the MCGRP. In Section 4.2 we explain how some of these valid inequalities are used in our algorithm. It is noted that these valid inequalities used in our algorithm were also used for the MCGRP in [15] and [30].

3.4. Combined Model

In this section, we present our combined model which is the model used in our Branch-and-Cut-and-Price algorithm. The purpose is to construct a model where valid inequalities can be added as cuts to the model without changing the structure of the pricing problem used to generate columns. The combined model is inspired by the modeling technique in [23] for the CVRP.

We use two types of variables in our combined model. From (IP Master), we use the route variables:

$$\lambda^r = \begin{cases} 1 & \text{if route } r \text{ is used in the solution} \\ 0 & \text{otherwise} \end{cases}$$

From the aggregated model, we use the aggregated deadheading variables:

$$x_e = \text{total number of times edge } e \text{ is deadheaded}$$

$$y_a = \text{total number of times arc } a \text{ is deadheaded}$$

In a standard Set Partitioning model, the pricing problem, defined by (IP Pricing), returns a vector containing the values of $([h_i^r], [s_e^r], [t_a^r])$ along with the cost c_r. In our combined model, we let the pricing problem, defined by (IP Pricing), return a vector containing the values of $([u_e^r], [w_a^r], [h_i^r], [s_e^r], [t_a^r])$ along with the cost c_r where

$$u_e^r = \text{number of times route } r \text{ deadheads edge } e$$

$$w_a^r = \text{number of times route } r \text{ deadheads arc } a$$

and h_i^r, s_e^r, and t_a^r are defined as in Section 3.2.

With this vector, we can state a combined model for the MCGRP where constraints (8)–(12) are identical to the complete set of constraints in (IP Master), i.e., constraints (1)–(5). Constraints (6) and (7) connect the route variables λ^r to the aggregated deadheading variables x_e and y_a.

(IP Combined)

$$\text{minimize} \quad \sum_{e \in E} c_e x_e + \sum_{a \in A} c_a y_a$$

$$\text{s.t.} \quad \sum_{r \in \Omega} u_e^r \lambda^r - x_e = 0, \forall e \in E \tag{6}$$

$$\sum_{r \in \Omega} w_a^r \lambda^r - y_a = 0, \forall a \in A \tag{7}$$

$$\sum_{r \in \Omega} h_i^r \lambda^r = 1, \forall i \in N_R \tag{8}$$

$$\sum_{r \in \Omega} s_e^r \lambda^r = 1, \forall e \in E_R \tag{9}$$

$$\sum_{r \in \Omega} t_a^r \lambda^r = 1, \forall a \in A_R \tag{10}$$

$$\sum_{r \in \Omega} \lambda^r \leq |K| \tag{11}$$

$$\lambda^r \geq 0 \text{ and integer } \forall r \in \Omega \tag{12}$$

$$x_e \geq 0 \text{ and integer } \forall e \in E \tag{13}$$

$$y_a \geq 0 \text{ and integer } \forall a \in A \tag{14}$$

Figure 1 displays the structure of (IP Combined). The structure shows that the pricing problem is only affected by the top part of the model and remains unaffected by valid inequalities added to the model as cuts using only the aggregated variables x_e and y_a. Therefore, the valid inequalities from [10] which are based on aggregated variables can, when extended to the MCGRP, be added to (IP Combined) without affecting the structure of the pricing problem used for column generation.

For the LP relaxation of (IP Combined), let $\phi = [\phi_e]$ and $\psi = [\psi_a]$ be the vectors of dual variables corresponding to

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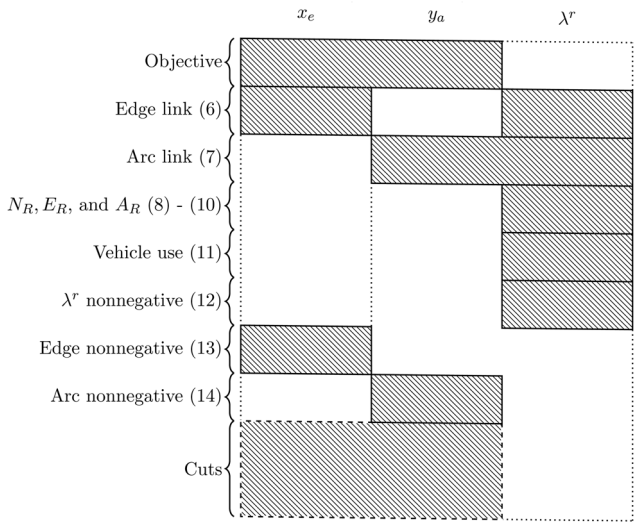


FIG. 1. Structure of (IP Combined).

constraints (6) and (7). As for (IP Master), we let $\pi = [\pi_i]$, $\mu = [\mu_e]$, and $\rho = [\rho_a]$ be the vectors of dual variables corresponding to constraints (8), (9), and (10), respectively. Finally, let σ be the dual variable corresponding to constraint (11).

Due to the linking constraints (6) and (7), the objective function of the pricing problem changes slightly compared to (IP Pricing). The traversal costs are now associated with the x_e and y_a variables and thus not directly included in the pricing problem. They will, however affect the pricing problem indirectly through the dual variables. The Pricing Problem is thus defined as follows:

(IP C-Pricing) Consider a graph with costs $-\pi_i$, $-\mu_e$, and $-\rho_a$ for servicing nodes i, edges e, and arcs a, respectively, and costs $-\phi_e$ and $-\psi_a$ for deadheading edges and arcs, respectively. Determine in this graph the minimum cost elementary path, starting and ending at the depot, subject to the vehicle capacity constraint.

As for (IP Pricing), the cost of this path exceeds by only a constant (σ) the reduced cost of its corresponding column in (IP Combined).

3.5. Valid Inequalities

The *Capacity constraints* from [10] ensure that for any set $S \subseteq N'$, a sufficient number of vehicles enters and leaves S. For the MCGRP, the capacity constraints are given by:

$$\sum_{e \in \delta^u(S)} x_e + \sum_{a \in \delta^+(S) \cup \delta^-(S)} y_a \geq 2\gamma(S) - |\delta_R(S)| \, \forall S \subseteq N' \tag{15}$$

The *Odd Edge Cutset constraints* from [10] ensure that for any set S, if the cutset contains an odd number of required links, then at least one deadheading will be performed. For the MCGRP, the Odd Edge Cutset constraint is:

$$\sum_{e \in \delta^u(S)} x_e + \sum_{a \in \delta^+(S) \cup \delta^-(S)} y_a \geq 1 \, \forall S \subseteq N' \text{ with } |\delta_R(S)| \text{odd} \tag{16}$$

In some cases, (16) is violated while (15) is respected, and in other cases, the opposite is true. This shows that both inequalities are useful.

Other valid inequalities for the CARP can also be converted to the MCGRP. In [3] we showed how the *Disjoint path inequalities #1 (DP1)* from [10] can be converted to the MCGRP.

4. BRANCH-AND-CUT-AND-PRICE ALGORITHM

This section describes our Branch-and-Cut-and-Price algorithm in detail. In Section 4.1, we describe how we generate the columns in the initial LP. In Section 4.2, we describe initial constraints that are added only at the root of our branching tree, and in Section 4.3, we describe separation and constraint tightening techniques that are applied throughout the branching tree. Because our procedure for identifying columns with negative reduced cost is strongly affected by our branching decision, we describe our branching strategy in Section 4.4, before describing our procedure for generating columns in Section 4.5. Finally, in Section 4.6, we describe stabilization techniques for our algorithm.

4.1. Generation of Initial Columns

For initializing the LP, we generate a dummy route for each required node, arc, and edge, servicing only that particular entity. In addition, we run a simple savings algorithm in which we repeatedly merge entities into feasible routes, until no further merges can feasibly be made. Each route generated during this merging process is also added to the LP.

4.2. Initial Cut Generation

In this section, we describe the initial generation of cuts (15) and (16), which is performed after the initial set of columns has been generated, but before solving the root node. Constraints are initially added at the root node in two ways.

We first consider sets that consist of a single node $S = \{i\}$. If $Q(S)$ exceeds the vehicle capacity, the Capacity cuts (15) can strengthen the model, hence we add the cuts. If an odd number of required links are adjacent to S, the Odd Edge Cutsets (16) are added. Our computational experiments indicate that both types of cuts help improve the run time of our algorithm.

Next we consider mutually disjoint cutsets $(U_h, N \setminus U_h)$ with $U_1 = \{1\}$ and $U_1 \subset U_2 \subset U_3 \subset U_4 \cdots \subset N$. Furthermore, in-between considering U_h and U_{h+1} we iteratively add a node $i \in \{U_{h+1} \setminus U_h\}$ to U_h and consider the resulting cutset. The nodes in $U_{h+1} \setminus U_h$ are included in arbitrary order. For

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each cut set U, we consider in turn the nodes of the connected components of $G(N \setminus U)$ as the set S. We add constraints (15) and (16) based on S if they are relevant, i.e, if $Q(S)$ exceeds the vehicle capacity, or $|\delta_R(S)|$ is odd, respectively.

4.3. Separation Algorithms

In every node of the branching tree we try to identify violated capacity inequalities. These cuts are heuristically separated using a greedy method. The algorithm is an extension of a similar separation algorithm presented in [38] for the CVRP.

The algorithm uses every node as a seed and then iteratively expands the node set with the most promising neighboring node, which is the one that minimizes the slack of constraints (15). When expanding the set we avoid choosing any node that will lead to a previously considered set. This continues until no further nodes can be added to the set.

In any integer solution, the degree on all nodes must be even. The Odd Edge Cutset constraints (16) are introduced to address this. This is however a relatively weak lower bound for the degree of each node. The capacity constraints given by (15) do in some cases strengthen this bound. In constraints (17) a slightly altered version of constraints (15) and (16) is introduced.

$$\sum_{e \in \delta^u(i)} x_e + \sum_{a \in \delta^+(i) \cup \delta^-(i)} y_a \geq b_i, \forall i \in N \qquad (17)$$

In (17) we consider only single nodes i rather than general sets of nodes S as in (15) and (16). The right hand side b_i in (17) is initially equal to the tightest bound from (15) and (16) for $S = \{i\}$. Subsequently, for each node $i \in N$ we solve a version of (IP Combined) with the objective of minimizing the left hand side of (17), and we then set b_i to this value rounded up to the nearest even integer. We apply this method after the root node is solved and the capacity cuts have been separated. The algorithm is applied again at later stages in the Branch-and-Cut-and-Price algorithm as well. If an upper bound is known, a constraint is added to the model constraining the original objective in (IP Combined), which may strengthen the cut. Whenever the upper bound is improved we also reapply the method to further tighten the constraints.

In [14], a branching scheme based on the degree of each node is presented. Our approach can be seen as identifying the branching decisions in which one branch is infeasible and then adding the alternative branching to the model.

4.4. Branching

Branch-and-Price algorithms for routing problems often apply the branching strategy known as follow-on branching. The idea is that, for a pair of required entities $a, b \in N_R \cup E_R \cup A_R$, two branches are created in the branching tree. In the left branch (1-branch, also written as $a \rightarrow b$ in the following), the connection between a and b is fixed, i.e., a vehicle that services a must service b immediately after, and similarly, a vehicle that services b must have serviced a immediately before. In the right branch (0-branch, also written as $a \not\rightarrow b$ in the following) this connection is forbidden, i.e., b cannot be serviced immediately after a.

In our Branch-and-Cut-and-Price algorithm, we branch on follow-on in a slightly different way, because we work directly on the original sparse graph as suggested in [35]. A branching scheme for follow-on branching on this sparse graph for the CARP is suggested in [14]. We use a similar follow-on branching scheme. The details on how the follow-on branching is handled in the pricing algorithm is described in Section 4.5.

We consider two branches. In the 1-branch we fix the connection between a and b such that a vehicle must service a immediately before b, and furthermore we require that any deadheading between a and b occurs along a shortest path using the original costs, regardless of the values of the dual variables at a specific stage of the algorithm. We emphasize that the algorithm may be restricted to require deadheading along shortest paths because this will always be the case in an optimal solution. All shortest paths can therefore be calculated a priori. This branch also includes the situation where no deadheading between a and b is needed because they are connected. In the 0-branch, any solution where b follows immediately after a is forbidden.

We use a depth first strategy in the branching tree where the 1-branch is prioritized.

4.5. Column Generation

In the (IP Combined) model the route variables are not included from the start, but are generated dynamically. To generate the route variables (columns) we solve a Shortest Path Problem with Resource Constraints (SPPRC) using a labeling algorithm. In our implementation, we do not require paths to be elementary and therefore h_i^r, s_e^r, and t_a^r are permitted to take values larger than one. Accordingly, h_i^r, s_e^r, and t_a^r represent the number of times route r services node i, edge e, and arc a, respectively. See [22, 29] for more on the SPPRC, and for an implementation in a column generation setup, see [31]. As in [14, 35], we work directly on the original graph G which is sparse for the MCGRP. To solve the SPPRC we require the dual variables $\phi < 0$ and $\psi < 0$ to avoid infinite deadheading cycles. This is further explained in Section 4.6.

All traversal costs in our model are associated with the x_e and y_a variables, i.e., the route variables in (IP Combined) have cost coefficients of zero. Therefore, the graph we use for column generation has costs $-\phi_e$ and $-\psi_a$ for deadheading edges and arcs, respectively, and a cost of $-\pi_i$, $-\mu_e$, and $-\rho_a$ for servicing nodes i, edges e, and arcs a, respectively. We define $SP(i, j)$ to be the set of links on a shortest path from i to j in the original network with respect to the original traversal costs.

Each label L has three attributes: the accumulated cost $C(L)$, the vehicle's accumulated demand $AccDem(L)$, and the non-follower attribute $NF(L)$. The accumulated demand is initially zero and is increased by the demands of the services along the path from the depot. The non-follower attribute is

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the set of entities that we cannot service next. Label L_a dominates L_b if $C(L_a) \leq C(L_b)$, $AccDem(L_a) \leq AccDem(L_b)$ and $NF(L_a) \subseteq NF(L_b)$, which also includes the special case where $NF(L_a) = \emptyset$. We will further explain the non-follower attribute in the following section.

Let $i \in N$ and let L_i be a label on i, i.e., L_i represents a path starting at the depot and ending at node i. We extend L_i as follows. For all $a = (i,j) \in \delta^+(i) \cap \{A \setminus A_R\}$ we create a deadhead extension of L_i to j, i.e., where the arc a is deadheaded. For all $a = (i,j) \in \delta_R^+(i)$ we create a deadheading extension of L_i to j and if it is feasible with respect to capacity, we also create a service extension of L_i to j. In the latter, the arc a is serviced. For edges in $\delta(i)$, we proceed in a similar way. For any of the above extensions of edges and arcs, if node $j \in N_R$, i.e., the node that we extend to is required, we create a copy of the extended labels where we further service j if this is feasible with respect to capacity. As such, there is a maximum of four new labels created from L_i on node j for each edge or arc leaving node i.

In our pricing problem we must handle the branching rules described in Section 4.4. We first describe how the individual follow-on branchings are handled and then we describe the handling of multiple interacting branching rules.

We first consider the 0-branch rule forbidding the service of a required task $b \in N_R \cup E_R \cup A_R$ to follow immediately after another required task $a \in N_R \cup E_R \cup A_R$. For each required task a, we introduce a non-follower attribute NF_a which is initialized to $NF_a = \emptyset$ in the root of the branching tree. For the branching rule $a \not\to b$ we set $NF_a = \{b\}$.

Consider now a 1-branch rule requiring entity b to follow immediately after entity a. We will first assume that both entities are arcs. Let $a = (i_a, j_a) \in A_R$ and $b = (i_b, j_b) \in A_R$. We insert a new required arc (i_a, j_b) with demand equal to $q_a + q_b$ and cost equal to $\rho_a + \sum_{e \in SP(j_a, i_b)} \phi_e + \sum_{a \in SP(j_a, i_b)} \psi_a + \rho_b$. Furthermore, we deactivate the arcs a and b as required arcs, but keep them as deadheading arcs.

We follow the same procedure for any combination of required arcs/nodes and insert an arc representing them in the only way that the tasks can be serviced.

If both a and b are edges, four new arcs with capacity $q_a + q_b$ are created to handle the flexibility of traversal of edges: (i_a, j_b), (i_a, i_b), (j_a, j_b), and (j_a, i_b). The cost of arc (i_a, j_b) is $\mu_a + \sum_{e \in SP(j_a, i_b)} \phi_e + \sum_{a \in SP(j_a, i_b)} \psi_a + \mu_b$, and the cost of the other arcs is defined similarly. Again, the two original edges a and b are deactivated as required edges but kept as deadheading edges. In case only a or b is an edge, only two of these new arcs are created.

As the algorithm proceeds, several such 1-branch rules may result in several of the created arcs needing to be concatenated. If a, b, and c are all tasks and 1-branch rules $a \to b$ and $b \to c$ are in effect, they must be concatenated. We insert a new required arc (i_a, j_c) with demand equal to $q_a + q_b + q_c$ and cost calculated in the same way as when managing a single rule. As for the case with two tasks, we disable all other required tasks involving the tasks being merged. This can easily be extended to cases containing four or more tasks.

When there are edges among the tasks being concatenated, we use the dynamic programming approach presented in [42] for finding a shortest path through such edges while taking the different orientations into account. This method results in at most four new arcs independently of the number of connections being concatenated.

Having considered the interaction of multiple 1-branch rules, let us now consider any combination of 0-branch rules. Here, if entities a, b, and c are in the branch rules $a \not\to b$ and $b \not\to c$, then $NF_a = \{b\}$ and $NF_b = \{c\}$. If both rules $a \not\to b$ and $a \not\to c$ apply, then $NF_a = \{b, c\}$.

Finally, we address the cases where 1-branch and 0-branch rules interact. For branch rules $a \to b$, $b \not\to c$, and $c \not\to d$, no concatenation is necessary. The new arc representing $a \to b$ will be given a $NF_b = \{c\}$, and $c \not\to d$ will be unaffected. Thus with regard to interactions all 1-branches are first concatenated and then the relevant 0-branches are applied.

The weakness of this approach to branching is the 0-branches. When the size of the non-follower attributes is greater than one, they become more complex to dominate and thus affect the speed of the algorithm.

4.6. *Stabilization*

In some cases, the dual variables take undesirable values which negatively impact the structure of the pricing problem. Bounding the values of the dual variables can reduce this problem.

In (IP Combined), the values of ϕ and ψ can vary freely because the corresponding constraints are equality constraints. In the pure model as stated in (IP Combined), the values of these dual variables can be kept non-positive by stabilization, but cuts are added to the model and these may cause dual variables to obtain positive values. Assume that e is an edge for which $\phi_e > 0$. This means that the contribution of the deadheading variable x_e to the objective function is $-\phi_e < 0$. Hence, the pricing algorithm can potentially produce columns of cost $-\infty$ by repeated deadheading of e. We relax constraints (6) and (7) to $\leq$. This forces $\phi_e \leq 0 \, \forall e \in E$ and $\psi_a \leq 0 \, \forall a \in A$ and stabilizes our model by ensuring that the above situation is avoided.

Intuitively, this relaxation means that we can pay for deadheading an edge e (or an arc) by increasing the value of x_e without using the edge in a route variable λ_r. We say that we pay the price c_e for not having to use e in a route. Note that the very purpose of our valid inequalities is to force the use of deadheading variables. It is important to note that the pricing problem (IP C-Pricing) ensures that every route generated is connected. Therefore, in any integer solution, constraints (6) and (7) will hold with equality and we pay for exactly those edges (and arcs) used in the route.

In [14], the authors discuss an alternative approach with the implementation of simple cycle variables in the master problem, to address the issue of infinite cycles for the CARP. Their approach is based on dual-optimal inequalities from [11]. In their approach it is necessary to ensure non-negative

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reduced cost for deadheading on edges $e \in E$. The variables z_e represent the simple deadheading cycles (e, e), i.e., nodes (i, j, i). The cost on this cycle is $2c_e$, twice the cost of deadheading e. The addition of these variables to the master problem forms dual inequalities $\beta_e \leq c_e, \forall e \in E$, where β_e is the sum of all dual variables related to edge e. This has the effect of restricting the reduced cost of deadheading $\tilde{c}_e$ to $\tilde{c}_e \geq 0$, which corresponds to our approach where $\phi_e \leq 0$ and forces the reduced cost of the edge to be non-negative. The comparison of the approaches is not as straightforward with respect to simple tours for arcs, as such tours require traversal of multiple arcs/edges.

In our approach we have $\phi_e \leq 0$, $\forall e \in E$ and $\psi_a \leq 0$, $\forall a \in A$ which guarantee that there are no negative reduced cost cycles.

5. COMPUTATIONAL RESULTS

We have tested our algorithm on a large number of benchmark instances as shown in Tables 2–15 of Appendix. All computations are done on an HP EliteBook with an Intel Core 2 Duo CPU P8700 @2.53 GHz and 4 GB of RAM. The algorithm is implemented in C++ as a sequential algorithm, and we use IBM ILOG CPLEX 12.5.1 to solve the LP relaxations.

The information in Tables 2–15 is organized as follows. The first column gives the instance name. The next three columns show the best known upper and lower bounds, including the results in this paper, together with the corresponding gap, where an '*' is used to represent proven optimality. We have marked these best known bounds with a letter indicating the corresponding reference. We use *A* for [40], *B* for [32], *C* for [28], *D* for [2], *E* for [15], *F* for [24], *G* for [30], *H* for [21], *I* for [3], *J* for this paper, and *K* for [16].

The gap reported in [15] and [30] is the gap from CPLEX. In these two articles, no lower bound is provided if optimality is not reached, except for what could be inferred from the CPLEX gap. For both [15] and [30], the value that we use as their lower bound is the upper bound if optimality is reached and the reported root value otherwise. To highlight this, we use Gap' rather than Gap as header for the gap columns in Tables 2–15 for these two references.

The remaining columns show the results from six sources listed in chronological order from left to right, namely [3, 15, 16, 21, 30] and this paper (denoted by BLW), respectively. In fact, [15] and [30] presented the other exact algorithms for the problem, except for our earlier version of this algorithm which was presented in [3]. For [15] and [30] we report their upper bound, root value, the CPLEX gap, and the run time as reported in the papers. As [16] and [21] use heuristic methods we only report their upper bounds.

For comparison of run times, which are specified in seconds, both [15] and [30] report results computed using a PC equipped with 2 Intel Xeon Quad Core CPUs @3.0 GHz with 6 GB RAM. The run times reported in [3] were obtained on the same PC as the one used in BLW.

TABLE 1. Count of new best known values obtained in this paper

Instance class	#Instances	UB	LB	#Closed
CBMIX	23	3	21	3
BHW	20	0	12	5
MGGDB_.25	23	0	2	21
MGGDB_.30	23	0	1	22
MGGDB_.35	23	0	2	23
MGGDB_.40	23	0	3	23
MGGDB_.45	23	0	2	23
MGGDB_.50	23	0	4	22
MGVAL_.25	34	1	3	4
MGVAL_.30	34	3	4	1
MGVAL_.35	34	2	3	3
MGVAL_.40	34	0	18	3
MGVAL_.45	34	1	23	4
MGVAL_.50	34	2	19	2
Total	385	12	117	159

A lower or upper bound is marked in bold if the result was a new best known bound at the time when it was obtained. For [15] and [30], we have marked the root value in bold if it provides a new best known lower bound.

Across the columns, an '*' is marked in bold if the particular algorithm is the first to generate a pair of equal lower and upper bounds and thereby the first to prove optimality by itself. For example, in Table 3, our algorithm generates the same value of 502 as both lower and upper bound. This value was previously known as a lower bound, and also as an upper bound, but these two bounds were not produced by the same algorithm. Our algorithm is the first algorithm which produces the optimal value as both lower and upper bound, which leads to the '*' being marked in bold.

For both [3] and BLW, we report the performance by showing the upper and lower bounds, the objective value of the LP relaxation at the root node of the branching tree, the gap calculated as $100 \times (UB - LB)/((UB + LB)/2)$, and the total computing time in seconds, where '‡' ('†') indicates that we reached a time limit of 3 (6) hours. For BLW, we finally also show the number of generated nodes in the branching tree.

We use a '-' to indicate that no results were reported if an attempt was made to solve the instance, and a blank entry if no attempt to solve the instance was reported.

A summary of the number of new best known upper and lower bounds, and number of instances solved to optimality, across all 385 instances, can be found in Table 1.

Table 2 shows the results obtained for the set of 23 CBMix instances provided in [40]. These instances are the oldest and have therefore been used in several papers, in particular with regards to upper bounds. We solve 3 instances to proven optimality, including one instance solved to proven optimality for the first time. We improve the best known upper bound for three instances and the best known lower bound for 21 instances.

The BHW instances [2] were presented later than the CBMix instances and have thus been used in fewer papers. Table 3 shows the results obtained for these 20 instances.

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We prove optimality of 5 instances, and in 12 of the remaining instances we improve the lower bound. The DI-NEARP instances, which were also presented in [2], are larger than the other instances and have not been used in our tests.

In [15], the authors present a set of 138 Mggdb instances divided into 6 subsets. Results for these instances are presented in Tables 4–9. Out of the 138 instances, we prove optimality for 134 instances. We improve the best known lower bounds on all 14 previously unsolved instances. For many of the instances solved to optimality by both our algorithm and the ones in [15] and [30], our computation time is the smallest of the three. However, there are also instances where the other algorithms are faster than ours. Out of the 138 instances, 24 instances are not solved in [15] because the number of required vehicles is larger than their algorithm can handle.

The authors of [15] also present a set of 204 Mgval instances divided into 6 subsets. Results for these instances are presented in Tables 10–15. We prove optimality for 17 out of these 204 instances, and 6 of these were solved to proven optimality for the first time. We improve the lower bound for 70 instances and the upper bound for 9 instances. In relation to the high number of improved lower bounds for the Mgval instances, it is fair to add that our algorithm did not have much competition as far as the instances in Tables 13–15 are concerned, where the majority of the 70 improved lower bounds were found. In any case, our results can serve as a useful basis for comparison in relation to assessing the performance of future algorithms.

6. CONCLUSION AND FUTURE RESEARCH

We have presented the first Branch-and-Cut-and-Price algorithm for the MCGRP, in which we use variables representing routes as well as variables representing deadheading traversals of arcs and edges.

Based on extensive computational experiments, we have compared our algorithm with the other existing exact algorithms for the problem. The main conclusion is that our algorithm in many cases performs better than the other algorithms. In particular, our algorithm is stronger for problems with many vehicles, whereas the algorithm in [15], being based on a three-index model, is more sensitive to the number of vehicles. For the instances where our algorithm does not prove optimality within the time limit, the obtained lower bound is in many cases very strong, and we improve several best known lower bounds.

However, we should add that our algorithm has a relatively weak performance on the particular class of Mgval instances. Indeed, on some of these instances our algorithm ends with quite large gaps, and on the Mgval instances in general the algorithm in [30] is competitive to our algorithm.

In relation to future research, possible improvements to our algorithm could be made with respect to producing upper bounds. In particular, an obvious improvement is to implement a construction heuristic to be used for initialization of the algorithm. For comparison, [15] has a strong focus on obtaining a good initial solution. Furthermore, it could be an advantage to use a metaheuristic to run in parallel to the exact algorithm and simultaneously update the best known upper bound while also adding columns to the exact algorithm. Our algorithm is only initialized with routes produced by a simple construction heuristic. As a result, for some of the difficult instances, our algorithm does not obtain a feasible upper bound, which increases its run time.

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APPENDIX : DETAILED COMPUTATIONAL RESULTS

TABLE 2. Computational results for CBMix

	Best known			[15]				[3]					[16]	[30]				[21]	BLW (This paper)					
Instance	UB	LB	Gap	UB	Root	Gap'	Time	UB	LB	Root	Gap	Time	UB	UB	Root	Gap'	Time	UB	UB	LB	Root	Gap	Time	Nodes
CBMix1	2569^{I}	2569^{J}		**2587**	-	-	-	**2569**	**2547**	2516.5	0.9	†	2587					2585	2569	**2569**	2533.4	*	16771.1	25802
CBMix2	11809^{H}	11505^{J}	2.6					-	**11487**	11486.3	-	†	12241					**11809**	12236	**11505**	11504.3	6.2	†	642
CBMix3	3614^{H}	3540^{J}	2.1					3684	**3514**	3500.2	4.7	†	3643					**3614**	3634	**3540**	3531.7	2.6	†	12802
CBMix4	7511^{H}	7323^{J}	2.5					7582	**7300**	7249.6	3.8	†	7583					**7511**	7558	**7323**	7257.6	3.2	†	15388
CBMix5	4459^{H}	4396^{J}	1.4					5548	**4387**	4379.6	23.4	†	4531					**4459**	4459	**4396**	4380.3	1.4	†	19448
CBMix6	6968^{K}	6741^{J}	3.3					7643	**6738**	6737.4	12.6	†	**6968**					6969	7078	**6741**	6740.9	4.9	†	310
CBMix7	9444^{J}	9074^{J}	4.0					-	**9046**	9043.2	-	†	9859					**9461**	**9444**	**9074**	9073.3	4.0	†	166
CBMix8	10318^{H}	9993^{J}	3.2					12114	**9976**	9975.7	19.4	†	**10658**					**10318**	10768	**9993**	9984.4	7.5	†	3630
CBMix9	4002^{H}	3890^{J}	2.8					4044	**3837**	3789.6	5.3	†	4060					**4002**	4003	**3890**	3819.3	2.9	†	69650
CBMix10	7500^{H}	7355^{J}	2.0					7614	**7343**	7324.0	3.6	†	7755					**7500**	7548	**7355**	7337.6	2.6	†	16176
CBMix11	4487^{H}	4374^{J}	2.6					-	**4318**	4317.3	-	†	4561					**4487**	4612	**4374**	4363.8	5.3	†	1494
CBMix12	3138^{E}	3138^{D}		**3138**	-	*	-	3138	3138	3138.0	*	2037.2	3138					3138	3138	3138	3138.0	*	493.9	126
CBMix13	8984^{H}	8744^{J}	2.7					-	**8681**	8678.5	-	†	**9110**					**8984**	9543	**8744**	8729.1	8.7	†	1978
CBMix14	8443^{H}	8262^{J}	2.2					-	**8205**	8204.5	-	†	8671					**8443**	8849	**8262**	8252.6	6.9	†	1150
CBMix15	8249^{H}	8019^{J}	2.8					8355	**8013**	7922.1	4.2	†	8359					**8249**	8386	**8019**	7922.1	4.5	†	11950
CBMix16	8633^{J}	8480^{J}	1.8					-	**8446**	8445.4	-	†	**8933**					**8714**	**8633**	**8480**	8479.2	1.8	†	2054
CBMix17	4034^{H}	4014^{J}	0.5					-	**3943**	3941.4	-	†	**4037**					**4034**	4034	**4014**	3954.1	0.5	†	24044
CBMix18	7044^{H}	6877^{J}	2.4					7137	**6856**	6845.6	4.0	†	7254					**7044**	7049	**6877**	6867.5	2.5	†	3122
CBMix19	16244^{H}	15658^{J}	3.7					-	**15628**	15621.3	-	†	**16554**					**16244**	17481	**15658**	15652.5	11.0	†	1624
CBMix20	4738^{J}	4713^{J}	0.5					5068	**4647**	4640.3	8.7	†	4885					**4794**	**4738**	**4713**	4680.7	0.5	†	11228
CBMix21	17889^{H}	17324^{J}	3.2					18201	**17295**	17259.7	5.1	†	18509					**17889**	18309	**17324**	17282.3	5.5	†	3806
CBMix22	1941^{A}	1936^{J}	0.3	1941	-	-	-	1941	**1905**	1896.0	1.9	†	1941					1941	1941	**1936**	1934.0	0.3	†	21288
CBMix23	780^{A}	780^{E}		780	-	*	-	780	716	677.3	8.6	†	780					780	780	780	780.0	*	2.7	0
Number of new best known values				2		2		1	21		0		6					18	3	21		1		

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TABLE 3. Computational results for BHW

	Best known			[15]				[3]					[16]	[30]				[21]	BLW (This paper)					
Instance	UB	LB	Gap	UB	Root	Gap'	Time	UB	LB	Root	Gap	Time	UB	UB	Root	Gap'	Time	UB	UB	LB	Root	Gap	Time	Nodes
BHW1	337^{C}	337^{I}						337	**337**	332.0	*	987.9						337	337	337	332.0	*	136.0	2434
BHW2	470^{C}	470^{D}						470	470	470.0	*	0.2						470	470	470	470.0	*	0.1	12
BHW3	415^{C}	415^{I}						415	**415**	401.6	*	29.8						415	415	415	401.9	*	4.0	294
BHW4	240^{C}	240^{D}						240	240	240.0	*	15.6						240	240	240	240.0	*	2.8	2
BHW5	502^{H}	502^{D}						-	500	500.0	-	†						**502**	502	502	502.0	*	12050.0	598
BHW6	388^{C}	388^{D}						-	385	385.0	-	†						388	395	385	385.0	2.6	†	3414
BHW7	1074^{H}	1047^{I}	2.5					-	**1047**	1046.3	-	†						**1074**	1117	1047	1046.3	6.5	†	966
BHW8	668^{H}	666^{J}	0.3					-	**665**	664.1	-	†						**668**	673	**666**	665.8	1.0	†	800
BHW9	875^{H}	858^{I}	2.0					-	**858**	857.4	-	†						**875**	876	858	857.4	2.1	†	3198
BHW10	8584^{H}	8370^{J}	2.5					-	**8310**	8304.2	-	†						**8584**	8710	**8370**	8361.9	4.0	†	8046
BHW11	4952^{H}	4720^{J}	4.8					-	**4690**	4689.5	-	†						**4952**	5317	**4720**	4710.7	11.9	†	14424
BHW12	10906^{H}	10763^{J}	1.3					-	**10605**	10603.2	-	†						**10906**	10969	**10763**	10690.1	1.9	†	3540
BHW13	14428^{H}	14193^{J}	1.6					-	**13952**	13948.4	-	†						**14428**	15001	**14193**	14141.9	5.5	†	2562
BHW14	24988^{H}	24405^{J}	2.4					-	**24377**	24375.3	-	†						**24988**	25064	**24405**	24387.4	2.7	†	22856
BHW15	15354^{H}	15194^{J}	1.0					16929	**15130**	15124.4	11.2	†						**15354**	15489	**15194**	15163.2	1.9	†	28318
BHW16	43567^{H}	42545^{J}	2.4					-	**42506**	42486.0	-	†						**43567**	43827	**42545**	42535.0	3.0	†	4338
BHW17	26116^{H}	25673^{J}	1.7					-	**25570**	25567.1	-	†						**26116**	26616	**25673**	25664.9	3.6	†	6488
BHW18	15318^{H}	14928^{J}	2.6					16774	**14840**	14837.9	12.2	†						**15318**	15791	**14928**	14909.7	5.6	†	11326
BHW19	9397^{H}	9282^{J}	1.2					10942	**9197**	9193.9	17.3	†						**9397**	9485	**9282**	9269.9	2.2	†	16682
BHW20	16162^{H}	15721^{J}	2.8					-	-	-	-	†						**16162**	19761	**15721**	15721.0	22.8	†	722
Number of new best known values								0	15		4							15	0	12		1		

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TABLE 4. Computational results for Mggdb {0.25}

	Best known			[15]				[3]					[16]	[30]				[21]	BLW (This paper)					
Instance	UB	LB	Gap	UB	Root	Gap'	Time	UB	LB	Root	Gap	Time	UB	UB	Root	Gap'	Time	UB	UB	LB	Root	Gap	Time	Nodes
mggdb_.25_1	280^{E}	280^{E}		**280**	251	*	79.4	280	280	275.8	*	57.8	280	280	277	*	30.4	280	280	280	280.0	*	0.3	12
mggdb_.25_2	349^{E}	349^{G}		**349**	**317**	5.6	†	352	**346**	341.8	1.7	‡	359	349	**349**	*	15.9	349	349	349	349.0	*	0.4	4
mggdb_.25_3	278^{E}	278^{E}		**278**	253.5	*	923.9	278	278	270.4	*	601.4	286	278	275	*	23.5	278	278	278	276.5	*	0.6	48
mggdb_.25_4	289^{E}	289^{E}		**289**	272	*	22.7	289	289	285.5	*	0.8	289	289	286	*	6.1	289	289	289	285.9	*	0.5	32
mggdb_.25_5	394^{E}	394^{I}		**394**	**364**	6.4	†	394	**394**	391.1	*	2.3	410	394	393	*	62.5	394	394	394	391.1	*	0.5	20
mggdb_.25_6	292^{E}	292^{E}		**292**	284	*	14.9	292	292	292.0	*	0.2	295	292	292	*	2.7	292	292	292	292.0	*	0.2	2
mggdb_.25_7	290^{E}	290^{E}		**290**	275	*	40.4	290	290	287.4	*	0.4	302	290	290	*	3.9	290	290	290	290.0	*	0.2	2
mggdb_.25_8	336^{I}	334^{J}	0.6					**336**	**333**	332.0	0.9	‡	351	358	332	7.3	†	336	336	**334**	332.0	0.6	‡	99446
mggdb_.25_9	309^{I}	309^{J}						**309**	**308**	303.4	0.3	‡	316	349	300	14.0	†	310	309	**309**	303.8	*	2514.5	14080
mggdb_.25_10	265^{E}	265^{E}		**265**	**265**	*	2.8	265	265	265.0	*	1.2	265	265	265	*	2.7	265	265	265	265.0	*	1.5	12
mggdb_.25_11	356^{E}	356^{G}		**356**	**345**	3.1	†	356	**349**	347.9	2.0	‡	369	356	**356**	*	230.6	356	356	353	351.0	0.8	‡	13704
mggdb_.25_12	459^{E}	459^{I}		**459**	**400**	3.7	†	459	**459**	454.5	*	4.9	465	459	439	*	616.0	459	459	459	454.6	*	1.4	14
mggdb_.25_13	388^{E}	388^{F}		**388**	**374**	3.4	†	388	388	385.6	*	41.4	392	388	376	*	5716.6	392	388	388	385.6	*	8.0	22
mggdb_.25_14	107^{E}	107^{E}		**107**	**107**	*	1.4	107	107	107.0	*	2.3	107	107	107	*	3.1	107	107	107	107.0	*	0.8	12
mggdb_.25_15	55^{E}	55^{E}		**55**	**55**	*	0.5	55	55	55.0	*	3.2	55	55	55	*	1.2	55	55	55	55.0	*	1.6	16
mggdb_.25_16	98^{E}	98^{E}		**98**	**98**	*	5.0	98	98	97.0	*	7121.4	98	98	98	*	7.3	98	98	98	97.0	*	1332.1	5154
mggdb_.25_17	71^{E}	71^{E}		**71**	**71**	*	1.0	71	71	71.0	*	23.4	71	71	71	*	1.8	71	71	71	71.0	*	3.5	22
mggdb_.25_18	144^{E}	144^{G}		**144**	**139**	3.5	†	144	**140**	139.7	2.8	‡	144	144	**144**	*	6.1	144	144	144	144.0	*	23.5	32
mggdb_.25_19	53^{E}	53^{E}		**53**	47	*	1.1	53	53	52.5	*	0.0	53	53	53	*	1.4	53	53	53	53.0	*	0.1	6
mggdb_.25_20	116^{E}	116^{E}		**116**	**116**	*	7.8	116	116	116.0	*	6.3	117	116	116	*	11.7	116	116	116	116.0	*	3.8	6
mggdb_.25_21	146^{E}	146^{I}		**146**	**145**	0.7	†	146	**146**	146.0	*	25.9	146	146	146	*	27.9	146	146	146	146.0	*	14.6	18
mggdb_.25_22	160^{I}	160^{F}						**160**	160	160.0	*	104.3	168	160	160	*	282.6	160	160	160	160.0	*	75.4	32
mggdb_.25_23	181^{I}	181^{F}						**181**	181	181.0	*	1008.2	186	181	181	*	19325.6	181	181	181	181.0	*	208.8	24
Number of new best known values				19		12		4	8		6		0	0		3		0	0	2		1		

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TABLE 5. Computational results for Mggdb {0.30}

	Best known			[15]				[3]					[16]	[30]				[21]	BLW (This paper)					
Instance	UB	LB	Gap	UB	Root	Gap'	Time	UB	LB	Root	Gap	Time	UB	UB	Root	Gap'	Time	UB	UB	LB	Root	Gap	Time	Nodes
mggdb_.30_1	273^{E}	273^{E}		**273**	238.25	*	4279.5	273	273	264.4	*	76.2	276	273	271	*	77.9	273	273	273	270.0	*	8.8	506
mggdb_.30_2	301^{E}	301^{I}		**301**	**270**	5.7	†	301	**301**	295.9	*	229.5	314	301	300	*	11631.5	301	301	301	300.5	*	0.6	12
mggdb_.30_3	270^{E}	270^{E}		**270**	253	*	4447.9	270	270	267.2	*	1.1	278	270	267	*	17.2	270	270	270	267.2	*	1.7	156
mggdb_.30_4	260^{E}	260^{E}		**260**	231	*	39.7	260	260	252.5	*	4.4	260	260	260	*	2.7	260	260	260	253.6	*	3.2	214
mggdb_.30_5	388^{E}	388^{I}		**388**	**369**	2.8	†	388	**388**	388.0	*	0.7	399	388	388	*	37.7	388	388	388	388.0	*	0.7	32
mggdb_.30_6	276^{E}	276^{E}		**276**	235	*	443.4	276	276	268.4	*	198.6	276	276	276	*	56.8	276	276	276	268.4	*	155.3	3178
mggdb_.30_7	273^{E}	273^{E}		**273**	228.5	*	2870.3	273	273	256.8	*	1553.9	277	273	272	*	810.1	273	273	273	256.8	*	1417.3	19982
mggdb_.30_8	331^{I}	331^{J}						**331**	328	321.6	0.9	‡	338	364	324	11.0	†	331	331	**331**	325.0	*	1093.9	6452
mggdb_.30_9	281^{I}	281^{I}						**281**	**281**	275.4	*	3435.3	284	317	273	13.9	†	281	281	281	276.4	*	300.9	1868
mggdb_.30_10	242^{E}	242^{E}		**242**	228	*	12.1	242	242	237.5	*	439.9	242	242	242	*	8.9	242	242	242	241.7	*	1.8	62
mggdb_.30_11	387^{E}	387^{G}		**387**	**381**	1.6	†	-	**382**	381.3	-	‡	399	387	**387**	*	250.8	387	387	382	381.3	1.3	‡	10302
mggdb_.30_12	467^{E}	467^{I}		**467**	**395**	6.4	†	467	**467**	461.6	*	2.0	472	467	452	*	1586.1	467	467	467	467.0	*	0.2	2
mggdb_.30_13	483^{I}	483^{I}		**486**	**447**	8.0	†	**483**	**483**	476.3	*	83.9	483	483	463	*	20676.0	483	483	483	476.3	*	9.4	104
mggdb_.30_14	101^{E}	101^{E}		**101**	98	*	184.7	101	101	100.0	*	1.9	101	101	101	*	2.5	101	101	101	101.0	*	1.0	0
mggdb_.30_15	44^{E}	44^{E}		**44**	**44**	*	0.5	44	44	44.0	*	2.3	44	44	44	*	0.8	44	44	44	44.0	*	1.1	8
mggdb_.30_16	105^{E}	105^{E}		**105**	**105**	*	2.8	105	105	105.0	*	11.7	107	105	105	*	30.6	105	105	105	105.0	*	8.6	6
mggdb_.30_17	65^{E}	65^{E}		**65**	**65**	*	0.6	65	65	65.0	*	10.3	67	65	65	*	1.4	65	65	65	65.0	*	3.6	10
mggdb_.30_18	144^{E}	144^{E}		**144**	**144**	*	1.6	144	144	144.0	*	217.7	144	144	144	*	7.0	144	144	144	144.0	*	25.0	38
mggdb_.30_19	51^{E}	51^{E}		**51**	50	*	0.4	51	51	51.0	*	0.2	51	51	51	*	1.0	51	51	51	51.0	*	0.1	4
mggdb_.30_20	94^{E}	94^{E}		**94**	91	*	27.3	94	94	91.8	*	10.7	97	94	93	*	24.3	94	94	94	93.2	*	5.6	36
mggdb_.30_21	121^{E}	121^{F}		**121**	**120**	0.8	†	121	121	120.5	*	89.5	122	121	121	*	24.9	121	121	121	121.0	*	13.2	4
mggdb_.30_22	153^{I}	153^{I}						**153**	**153**	152.2	*	374.4	156	153	153	*	13595.8	153	153	153	153.0	*	105.0	258
mggdb_.30_23	167^{I}	167^{F}						**167**	167	166.9	*	677.8	171	167	167	*	20184.6	171	167	167	167.0	*	229.1	50
Number of new best known values				19		13		5	7		8		0	0		1		0	0	1		1		

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TABLE 6. Computational results for Mggdb {0.35}

	Best known			[15]				[3]					[16]	[30]				[21]	BLW (This paper)					
Instance	UB	LB	Gap	UB	Root	Gap'	Time	UB	LB	Root	Gap	Time	UB	UB	Root	Gap'	Time	UB	UB	LB	Root	Gap	Time	Nodes
mggdb_.35_1	252^E	252^E		**252**	232	*	601.0	252	251	242.4	0.4	‡	252	252	252	*	41.9	252	252	252	247.5	*	78.6	978
mggdb_.35_2	284^E	284^E		**284**	271	*	1435.2	284	284	284.0	*	1.4	284	284	284	*	19.1	284	284	284	284.0	*	0.5	2
mggdb_.35_3	243^E	243^E		**243**	208	*	1701.0	243	243	235.8	*	185.7	243	243	243	*	109.0	243	243	243	238.5	*	10.2	398
mggdb_.35_4	242^E	242^E		**242**	180	*	27.6	242	242	235.0	*	3.7	242	242	237	*	14.1	242	242	242	235.0	*	1.6	142
mggdb_.35_5	309^E	309^I		**309**	**282**	6.1	†	309	**309**	302.7	*	37.5	317	309	308.5	*	1056.1	309	309	309	307.8	*	0.6	10
mggdb_.35_6	262^E	262^E		**262**	235	*	1552.4	262	262	246.8	*	1241.9	262	262	250	*	140.8	262	262	262	254.8	*	4.9	256
mggdb_.35_7	272^E	272^E		**272**	247	*	412.7	272	272	266.0	*	68.6	272	272	272	*	12.9	272	272	272	272.0	*	0.4	2
mggdb_.35_8	316^I	316^J						**316**	**315**	309.8	0.3	‡	321	337	303	10.1	†	320	316	**316**	312.6	*	205.0	2586
mggdb_.35_9	266^I	266^J						**266**	**265**	259.1	0.4	‡	274	292	260	11.0	†	267	266	**266**	261.2	*	801.8	3732
mggdb_.35_10	268^E	268^E		**268**	258	*	813.6	268	267	265.8	0.4	‡	268	268	268	*	6.3	268	268	268	265.8	*	3532.0	8692
mggdb_.35_11	303^E	303^E		**303**	293	*	3875.7	313	301	300.0	3.9	‡	313	303	303	*	255.6	303	303	303	303.0	*	69.5	106
mggdb_.35_12	461^E	461^E		**461**	369	*	5229.9	461	461	461.0	*	2.4	461	461	441	*	197.5	461	461	461	461.0	*	0.9	36
mggdb_.35_13	417^E	417^F		**417**	**402**	3.0	†	417	417	416.9	*	5.3	435	417	410	*	674.1	417	417	417	416.9	*	4.4	2
mggdb_.35_14	84^E	84^E		**84**	81	*	369.3	84	84	83.5	*	0.8	85	84	83	*	45.2	84	84	84	84.0	*	2.3	56
mggdb_.35_15	44^E	44^E		**44**	**44**	*	0.7	44	44	44.0	*	5.8	44	44	44	*	1.2	44	44	44	44.0	*	1.3	24
mggdb_.35_16	75^E	75^E		**75**	71.5	*	2271.4	75	75	73.2	*	193.9	75	75	73	*	9276.6	75	75	75	73.6	*	82.4	178
mggdb_.35_17	62^E	62^E		**62**	**62**	*	0.9	62	62	62.0	*	5.1	62	62	62	*	2.6	62	62	62	62.0	*	2.5	22
mggdb_.35_18	135^E	135^E		**135**	**135**	*	0.4	135	135	135.0	*	81.2	137	135	135	*	12.4	135	135	135	135.0	*	14.4	42
mggdb_.35_19	51^E	51^E		**51**	**51**	*	0.2	51	51	51.0	*	0.1	51	51	51	*	0.8	51	51	51	51.0	*	0.1	10
mggdb_.35_20	96^E	96^E		**96**	93	*	75.8	96	96	93.5	*	27.6	96	96	93	*	23.7	96	96	96	93.5	*	4.3	46
mggdb_.35_21	120^E	120^I		**120**	**117.5**	2.1	†	120	**120**	119.0	*	60.0	122	120	120	*	26.8	120	120	120	120.0	*	17.0	30
mggdb_.35_22	139^I	139^I						**139**	**139**	138.2	*	191.5	143	139	139	*	266.9	139	139	139	139.0	*	28.2	22
mggdb_.35_23	179^I	179^I						**179**	**179**	178.5	*	304.7	185	194	176	9.3	†	179	179	179	178.5	*	79.7	76
Number of new best known values				19		16		4	6		5		0	0		0		0	0	2		2		

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TABLE 7. Computational results for Mggdb {0.40}

	Best known			[15]				[3]					[16]	[30]				[21]	BLW (This paper)					
Instance	UB	LB	Gap	UB	Root	Gap'	Time	UB	LB	Root	Gap	Time	UB	UB	Root	Gap'	Time	UB	UB	LB	Root	Gap	Time	Nodes
mggdb_.40_1	279^{E}	279^{E}		**279**	248.33	*	392.9	279	279	269.9	*	93.0	279					279	279	279	271.5	*	14.6	706
mggdb_.40_2	308^{E}	308^{I}		**308**	**281**	1.9	†	308	**308**	297.8	*	570.7	320					308	308	308	307.0	*	1.1	22
mggdb_.40_3	225^{E}	225^{E}		**225**	208	*	88.2	225	225	221.1	*	3.7	229					225	225	225	223.3	*	0.4	16
mggdb_.40_4	238^{E}	238^{E}		**238**	205	*	9.8	238	238	235.0	*	0.2	238					238	238	238	238.0	*	0.2	0
mggdb_.40_5	344^{E}	344^{I}		**344**	**289**	7.0	†	344	**344**	339.9	*	2.8	346					344	344	344	339.9	*	2.0	90
mggdb_.40_6	270^{E}	270^{E}		**270**	247	*	418.9	270	270	266.4	*	10.4	281					270	270	270	270.0	*	0.3	4
mggdb_.40_7	282^{E}	282^{E}		**282**	215	*	7631.9	282	275	267.0	2.5	‡	283					282	282	282	279.6	*	2.7	332
mggdb_.40_8	331^{H}	331^{J}						333	**326**	320.4	2.1	‡	340					**331**	331	**331**	324.8	*	7133.8	60750
mggdb_.40_9	275^{I}	275^{J}						**275**	**273**	265.6	0.7	‡	285					275	275	**275**	268.9	*	6030.8	27676
mggdb_.40_10	191^{E}	191^{E}		**191**	181	*	3.5	191	191	188.0	*	137.1	191					191	191	191	191.0	*	1.2	38
mggdb_.40_11	283^{E}	283^{J}		**283**	**270**	2.6	†	294	277	275.8	6.0	‡	287					283	283	**283**	282.0	*	272.9	318
mggdb_.40_12	412^{E}	412^{E}		**412**	314	*	10608.8	412	412	399.8	*	3.6	412					412	412	412	412.0	*	0.5	0
mggdb_.40_13	405^{E}	405^{I}		**405**	**373**	7.3	†	405	**405**	394.7	*	177.6	405					405	405	405	394.7	*	9.4	72
mggdb_.40_14	62^{E}	62^{E}		**62**	58	*	76.0	62	62	62.0	*	0.5	62					62	62	62	62.0	*	0.6	0
mggdb_.40_15	37^{E}	37^{E}		**37**	**37**	*	0.2	37	37	37.0	*	2.2	37					37	37	37	37.0	*	0.7	4
mggdb_.40_16	84^{E}	84^{E}		**84**	**84**	*	0.7	84	84	84.0	*	25.9	84					84	84	84	84.0	*	5.2	16
mggdb_.40_17	65^{E}	65^{E}		**65**	**65**	*	0.5	65	65	65.0	*	3.2	65					65	65	65	65.0	*	2.1	10
mggdb_.40_18	119^{E}	119^{E}		**119**	114	*	45.3	119	118	118.0	0.8	‡	122					119	119	119	119.0	*	23.1	22
mggdb_.40_19	38^{E}	38^{E}		**38**	**38**	*	0.6	38	38	38.0	*	0.0	38					38	38	38	38.0	*	0.1	0
mggdb_.40_20	94^{E}	94^{E}		**94**	92	*	15.9	94	94	94.0	*	5.4	94					94	94	94	94.0	*	3.1	10
mggdb_.40_21	104^{E}	104^{E}		**104**	100.25	*	207.6	104	104	103.9	*	40.9	106					104	104	104	104.0	*	8.0	8
mggdb_.40_22	129^{F}	129^{F}						129	129	128.2	*	116.5	132					129	129	129	129.0	*	44.0	26
mggdb_.40_23	160^{I}	160^{F}						**160**	160	160.0	*	198.3	165					163	160	160	160.0	*	53.8	12
Number of new best known values				19		15		2	5		5		0					1	0	3		3		

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TABLE 8. Computational results for Mggdb {0.45}

	Best known			[15]				[3]					[16]	[30]				[21]	BLW (This paper)					
Instance	UB	LB	Gap	UB	Root	Gap'	Time	UB	LB	Root	Gap	Time	UB	UB	Root	Gap'	Time	UB	UB	LB	Root	Gap	Time	Nodes
mggdb_.45_1	259^E	259^E		**259**	211	*	283.4	259	259	256.5	*	2.0	259					259	259	259	258.0	*	0.3	26
mggdb_.45_2	298^E	298^I		**298**	**279**	4.8	†	298	**298**	291.6	*	403.9	302					298	298	298	298.0	*	0.3	4
mggdb_.45_3	237^E	237^E		**237**	210	*	246.1	237	237	235.4	*	0.7	245					237	237	237	237.0	*	0.1	0
mggdb_.45_4	228^E	228^E		**228**	186.33	*	10.3	228	228	222.3	*	1.3	228					228	228	228	226.5	*	0.2	4
mggdb_.45_5	350^E	350^I		**350**	**309**	4.4	†	350	**350**	346.4	*	4.0	357					350	350	350	350.0	*	0.3	14
mggdb_.45_6	218^E	218^E		**218**	182	*	109.0	218	218	215.8	*	1.5	225					218	218	218	217.7	*	0.4	36
mggdb_.45_7	243^E	243^E		**243**	171	*	121.2	243	243	224.6	*	968.3	243					243	243	243	243.0	*	0.3	4
mggdb_.45_8	296^I	296^F						**296**	296	295.2	*	3.5	312					296	296	296	295.8	*	3.0	18
mggdb_.45_9	277^I	277^I						**277**	**277**	269.4	*	7207.2	287					277	277	277	271.6	*	891.4	5390
mggdb_.45_10	214^E	214^E		**214**	196	*	84.9	214	214	212.2	*	30.6	214					214	214	214	213.1	*	2.3	64
mggdb_.45_11	297^E	297^J		**297**	**278**	4.4	†	301	**289**	287.7	4.1	‡	310					297	297	**297**	295.5	*	10055.7	12430
mggdb_.45_12	393^E	393^I		**393**	**321**	5.5	†	393	**393**	377.5	*	6.3	406					393	393	393	386.5	*	2.5	108
mggdb_.45_13	423^E	423^I		**423**	**371**	10.2	†	423	**423**	422.4	*	2.2	423					429	423	423	422.4	*	4.4	14
mggdb_.45_14	66^E	66^E		**66**	58	*	102.7	66	66	63.5	*	2.4	67					66	66	66	64.8	*	0.6	12
mggdb_.45_15	34^E	34^E		**34**	**34**	*	0.5	34	34	34.0	*	1.7	36					34	34	34	34.0	*	0.7	6
mggdb_.45_16	70^E	70^E		**70**	62	*	238.2	70	70	70.0	*	12.0	70					70	70	70	70.0	*	4.0	6
mggdb_.45_17	53^E	53^E		**53**	**53**	*	1.7	53	53	53.0	*	5.5	53					53	53	53	53.0	*	2.5	20
mggdb_.45_18	123^E	123^J		**123**	**112**	7.2	†	123	**121**	120.3	1.6	‡	123					123	123	**123**	123.0	*	20.0	24
mggdb_.45_19	48^E	48^E		**48**	40	*	0.9	48	48	48.0	*	0.0	48					48	48	48	48.0	*	0.1	4
mggdb_.45_20	78^E	78^E		**78**	76	*	7.2	78	78	77.3	*	2.3	78					78	78	78	78.0	*	2.0	6
mggdb_.45_21	122^E	122^E		**122**	120	*	10530.8	122	122	122.0	*	5.9	128					122	122	122	122.0	*	3.4	6
mggdb_.45_22	136^I	136^I						**136**	**136**	134.8	*	121.5	139					136	136	136	135.5	*	56.6	52
mggdb_.45_23	144^I	144^I						**144**	**144**	143.8	*	354.0	147					145	144	144	144.0	*	168.4	2
Number of new best known values				19		13		4	9		8		0					0	0	2		2		

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TABLE 9. Computational results for Mggdb {0.50}

	Best known			[15]				[3]					[16]	[30]				[21]	BLW (This paper)					
Instance	UB	LB	Gap	UB	Root	Gap'	Time	UB	LB	Root	Gap	Time	UB	UB	Root	Gap'	Time	UB	UB	LB	Root	Gap	Time	Nodes
mggdb_.50_1	214^E	214^E		**214**	178	*	86.8	214	214	207.3	*	7.5	214					214	214	214	214.0	*	0.3	2
mggdb_.50_2	269^E	269^I		**269**	**233**	6.8	†	269	**269**	259.5	*	40.7	281					269	269	269	265.5	*	3.5	138
mggdb_.50_3	218^E	218^E		**218**	184	*	399.7	218	218	205.6	*	22.4	218					218	218	218	217.6	*	0.2	4
mggdb_.50_4	219^E	219^E		**219**	172	*	16.4	219	219	214.8	*	0.5	219					219	219	219	219.0	*	0.2	2
mggdb_.50_5	292^E	292^E		**292**	226	*	2889.3	292	292	290.5	*	1.7	292					292	292	292	291.1	*	0.6	8
mggdb_.50_6	276^E	276^E		**276**	228	*	1171.8	276	276	270.9	*	9.4	276					276	276	276	274.9	*	0.9	58
mggdb_.50_7	265^E	265^E		**265**	237	*	379.0	265	265	258.7	*	15.7	274					265	265	265	264.5	*	0.4	24
mggdb_.50_8	310^I	310^J						**310**	**309**	299.1	0.3	‡	310					310	310	**310**	302.6	*	149.7	1744
mggdb_.50_9	265^I	265^J						**265**	259	255.7	2.3	‡	270					265	265	**265**	257.7	*	6036.7	35006
mggdb_.50_10	194^E	194^E		**194**	190	*	3.7	194	194	194.0	*	0.9	194					194	194	194	194.0	*	0.8	4
mggdb_.50_11	275^E	268^J	2.6	**275**	**249**	4.2	†	-	**266**	265.2	-	‡	278					275	275	**268**	265.5	2.6	‡	15406
mggdb_.50_12	445^E	445^E		**445**	352	*	13492.7	445	445	436.5	*	8.5	445					445	445	445	438.5	*	2.7	88
mggdb_.50_13	259^E	259^F		**259**	**214**	8.5	†	259	259	257.8	*	43.1	259					261	259	259	257.8	*	6.6	8
mggdb_.50_14	75^E	75^E		**75**	71	*	20.6	75	75	74.3	*	2.0	76					75	75	75	75.0	*	1.2	0
mggdb_.50_15	37^E	37^E		**37**	**37**	*	0.4	37	37	37.0	*	0.5	37					37	37	37	37.0	*	0.7	18
mggdb_.50_16	66^E	66^E		**66**	62	*	36.3	66	66	65.0	*	14.7	66					66	66	66	65.0	*	3.5	10
mggdb_.50_17	53^E	53^E		**53**	**53**	*	0.9	53	53	53.0	*	5.1	53					53	53	53	53.0	*	5.1	14
mggdb_.50_18	121^E	121^J		**121**	**111**	6.6	†	122	117	116.7	4.2	‡	122					121	121	**121**	121.0	*	47.4	24
mggdb_.50_19	44^E	44^E		**44**	39.74	*	0.3	44	44	44.0	*	0.0	44					44	44	44	44.0	*	0.1	2
mggdb_.50_20	81^E	81^E		**81**	75	*	20.7	81	81	79.0	*	11.0	81					81	81	81	79.0	*	1.3	16
mggdb_.50_21	86^E	86^E		**86**	80	*	15691.3	86	86	83.9	*	54.3	88					86	86	86	83.9	*	13.3	40
mggdb_.50_22	123^I	123^F						**123**	123	123.0	*	54.6	127					123	123	123	123.0	*	18.7	2
mggdb_.50_23	125^I	125^F						**125**	125	124.6	*	203.9	125					126	125	125	124.6	*	61.2	28
Number of new best known values				19		15		4	3		4		0					0	0	4		3		

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TABLE 10. Computational results for Mgval {0.25}

	Best known			[15]				[3]					[16]	[30]				[21]	BLW (This paper)					
Instance	UB	LB	Gap	UB	Root	Gap'	Time	UB	LB	Root	Gap	Time	UB	UB	Root	Gap'	Time	UB	UB	LB	Root	Gap	Time	Nodes
mgval_.25_1A	177^{E}	177^{E}		**177**	**177**	*	0.3						177	177	177	*	17.2	177	221	173	173.0	24.4	‡	2104
mgval_.25_1B	217^{E}	217^{E}		**217**	**217**	*	0.1						217	217	217	*	41.9	217	217	217	217.0	*	1992.2	2956
mgval_.25_1C	280^{J}	278^{G}	0.7										**335**	323	**278**	13.9	†	**292**	**280**	271	269.6	3.3	‡	38178
mgval_.25_2A	259^{E}	259^{E}		**259**	251	*	6.1						259	259	259	*	18.2	259	259	258	257.0	0.4	‡	3150
mgval_.25_2B	336^{E}	336^{E}		**336**	318	*	941.5						336	336	336	*	53.0	336	336	329	328.0	2.1	‡	4210
mgval_.25_2C	480^{H}	480^{J}											**528**	512	**479**	6.5	†	**480**	480	**480**	473.5	*	6442.6	22692
mgval_.25_3A	89^{E}	89^{E}		**89**	88	*	3.2						89	89	89	*	16.0	89	89	89	88.7	*	1628.4	570
mgval_.25_3B	125^{E}	125^{E}		**125**	112.5	*	7217.3						125	125	125	*	17.7	125	125	123	122.1	1.6	‡	19808
mgval_.25_3C	153^{E}	153^{G}		**153**	**129.5**	11.0	†						161	153	**153**	*	106.1	153	153	153	150.6	*	2567.5	19560
mgval_.25_4A	514^{E}	514^{G}		**514**	**504**	1.6	†						514	514	**514**	*	582.5	514	653	509	508.7	24.8	‡	824
mgval_.25_4B	537^{E}	537^{G}		**537**	**506.25**	4.7	†						541	537	**537**	*	1328.1	537	619	528	526.0	15.9	‡	1040
mgval_.25_4C	525^{E}	525^{G}		**525**	**504**	4.0	†						549	525	**525**	*	14925.2	525	535	523	522.4	2.3	‡	240
mgval_.25_4D	685^{H}	675^{G}	1.5										**724**	778	**675**	13.2	†	**685**	725	670	668.2	7.9	‡	3342
mgval_.25_5A	485^{E}	485^{E}		**485**	483	*	2,418.17						485	485	485	*	482.3	485	650	484	483.4	29.3	‡	308
mgval_.25_5B	493^{E}	493^{G}		**493**	**472**	4.3	†						500	493	**493**	*	918.2	493	576	484	484.0	17.4	‡	664
mgval_.25_5C	584^{E}	584^{G}		**584**	**561**	3.9	†						599	584	**584**	*	722.3	584	590	572	571.6	3.1	‡	804
mgval_.25_5D	645^{H}	635^{G}	1.6										**681**	741	**635**	14.3	†	**645**	647	634	633.1	2.0	‡	3230
mgval_.25_6A	274^{E}	274^{E}		**274**	**274**	*	11.8						274	274	274	*	47.9	274	289	266	266.0	8.3	‡	2258
mgval_.25_6B	263^{E}	263^{G}		**263**	**253.5**	1.9	†						263	263	257	*	11665.0	263	269	258	257.4	4.2	‡	3728
mgval_.25_6C	324^{H}	319^{J}	1.6										**337**	378	**316**	16.4	†	**324**	324	**319**	314.8	1.6	‡	54436
mgval_.25_7A	297^{E}	297^{E}		**297**	294.5	*	121.3						297	297	297	*	407.9	297	-	290	290.0	-	‡	1500
mgval_.25_7B	355^{E}	355^{G}		**355**	**351**	5.8	19.0						355	355	**355**	*	2153.2	355	433	353	352.7	20.4	‡	1422
mgval_.25_7C	380^{H}	374^{G}	1.6										**407**	437	**374**	14.4	†	**380**	389	371	370.2	4.7	‡	15054
mgval_.25_8A	510^{E}	510^{G}		**510**	**507**	0.4	†						510	510	**510**	*	384.1	510	540	508	508.0	6.1	‡	418
mgval_.25_8B	423^{E}	423^{G}		**423**	**405**	4.3	†						423	423	**423**	*	1910.6	423	464	420	419.3	10.0	‡	590
mgval_.25_8C	544^{H}	538^{G}	1.1										**591**	625	**538**	13.9	†	**544**	544	534	533.4	1.9	‡	9802
mgval_.25_9A	371^{E}	371^{G}		**371**	**367**	1.1	†						371	371	**371**	*	669.0	371	-	366	365.7	-	‡	280
mgval_.25_9B	358^{E}	358^{G}		**358**	**354**	1.1	†						363	358	**358**	*	20123.6	358	-	354	353.9	-	‡	298
mgval_.25_9C	365^{E}	361^{G}	1.1	**365**	**347**	4.9	†						369	365	**361**	1.1	†	368	449	354	353.7	23.7	‡	490
mgval_.25_9D	427^{H}	420^{J}	1.7										**478**	498	**418**	16.1	†	**427**	434	**420**	418.4	3.3	‡	2408
mgval_.25_10A	492^{E}	492^{E}		**492**	**492**	*	5.1						492	492	492	*	8200.5	492	-	491	491.0	-	‡	250
mgval_.25_10B	528^{E}	528^{E}		**528**	**528**	*	1.6						528	528	528	*	19187.6	528	636	528	528.0	18.6	‡	200
mgval_.25_10C	483^{E}	483^{G}		**483**	**480**	0.6	†						501	483	**483**	*	20129.2	483	-	478	477.7	-	‡	214
mgval_.25_10D	568^{H}	566^{G}	0.4										**616**	655	565.5	13.6	†	**568**	569	555	554.2	2.5	‡	2116
Number of new best known values				25		11							9	0		13		9	1	3		1		

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TABLE 11. Computational results for Mgval {0.30}

	Best known			[15]				[3]					[16]	[30]				[21]	BLW (This paper)					
Instance	UB	LB	Gap	UB	Root	Gap'	Time	UB	LB	Root	Gap	Time	UB	UB	Root	Gap'	Time	UB	UB	LB	Root	Gap	Time	Nodes
mgval_.30_1A	170^{E}	170^{E}		**170**	168	*	4.4						170	170	170	*	15.0	170	186	169	167.0	9.6	‡	1688
mgval_.30_1B	194^{E}	194^{E}		**194**	182	*	24.5						194	194	194	*	146.6	194	198	194	193.0	2.0	‡	11290
mgval_.30_1C	274^{J}	264^{J}	3.7										**280**	310	**255**	17.7	†	280	**274**	**264**	254.0	3.7	‡	71540
mgval_.30_2A	233^{E}	233^{E}		**233**	228.44	*	1.6						233	233	233	*	33.5	233	233	233	233.0	*	309.3	210
mgval_.30_2B	347^{E}	347^{E}		**347**	322.83	*	3310.0						347	347	347	*	140.7	347	352	340	339.5	3.5	‡	4934
mgval_.30_2C	495^{J}	492^{J}	0.6										**542**	534	**489**	8.4	†	**498**	**495**	**492**	482.1	0.6	‡	64170
mgval_.30_3A	105^{E}	105^{E}		**105**	103.5	*	6.3						105	105	105	*	73.9	105	106	104	104.0	1.9	‡	4022
mgval_.30_3B	115^{E}	115^{E}		**115**	108	*	31.3						115	115	115	*	51.1	115	115	114	113.6	0.9	‡	7518
mgval_.30_3C	153^{E}	151^{J}	1.3	**153**	**126.5**	12.1	†						156	153	**149**	2.6	†	153	153	**151**	148.4	1.3	‡	72160
mgval_.30_4A	477^{E}	477^{G}		**477**	**466**	1.8	†						477	477	**477**	*	2098.6	477	576	473	472.5	19.6	‡	438
mgval_.30_4B	533^{E}	531^{G}	0.4	**533**	**491**	5.2	†						537	533	**531**	0.4	†	533	617	524	523.5	16.3	‡	550
mgval_.30_4C	498^{E}	492^{G}	1.2	**498**	**468.5**	5.9	†						513	498	**492**	1.2	†	498	541	488	487.4	10.3	‡	1000
mgval_.30_4D	653^{H}	652^{G}	0.2										**718**	765	**652**	14.8	†	**653**	660	649	648.5	1.7	‡	3740
mgval_.30_5A	445^{E}	445^{E}		**445**	444	*	96.3						445	445	445	*	509.5	445	-	445	444.4	-	‡	252
mgval_.30_5B	490^{E}	484^{G}	1.2	**490**	**465**	4.6	†						492	490	**484**	1.2	†	492	516	480	479.7	7.2	‡	654
mgval_.30_5C	551^{E}	549^{G}	0.4	**551**	**513**	6.8	†						568	551	**549**	0.4	†	553	581	542	541.5	6.9	‡	818
mgval_.30_5D	618^{H}	612^{G}	1.0										**675**	736	**612**	16.9	†	**618**	628	605	604.5	3.7	‡	3328
mgval_.30_6A	252^{E}	252^{G}		**252**	**240**	1.8	†						252	252	**252**	*	139.8	252	286	240	239.7	17.5	‡	2148
mgval_.30_6B	262^{E}	262^{E}		**262**	251	*	6331.6						268	262	262	*	1425.9	263	263	255	254.1	3.1	‡	5868
mgval_.30_6C	317^{J}	311^{J}	1.9										**339**	364	**307**	15.7	†	**322**	**317**	**311**	307.3	1.9	‡	46678
mgval_.30_7A	324^{E}	324^{E}		**324**	**324**	*	1.7						324	324	324	*	3197.5	324	-	318	318.0	-	‡	1554
mgval_.30_7B	344^{E}	344^{G}		**344**	**336**	1.9	†						344	344	**344**	*	632.3	344	396	338	337.2	15.8	‡	2806
mgval_.30_7C	354^{H}	347^{G}	2.0										**380**	388	**347**	10.6	†	**354**	354	341	340.2	3.7	‡	16256
mgval_.30_8A	431^{E}	431^{G}		**431**	**427.5**	0.8	†						431	431	**431**	*	1108.1	431	535	426	426.0	22.7	‡	522
mgval_.30_8B	400^{E}	400^{G}		**400**	**390.5**	2.4	†						408	400	**400**	*	9578.2	400	419	396	395.3	5.6	‡	446
mgval_.30_8C	522^{H}	510^{G}	2.3										**570**	590	**510**	13.6	†	**522**	522	507	505.2	2.9	‡	12574
mgval_.30_9A	357^{E}	357^{G}		**357**	**356**	0.3	†						357	357	**357**	*	4549.3	357	-	352	352.0	-	‡	300
mgval_.30_9B	348^{E}	348^{G}		**348**	**344**	1.2	†						356	348	**348**	*	12667.8	348	-	342	341.4	-	‡	520
mgval_.30_9C	335^{E}	335^{G}		**335**	**330**	1.5	†						347	335	**335**	*	20185.9	335	408	334	334.0	19.9	‡	620
mgval_.30_9D	434^{H}	422^{G}	2.8										**475**	501	421.13	15.8	†	**434**	487	420	418.2	14.8	‡	2018
mgval_.30_10A	484^{E}	484^{G}		**484**	**481**	0.6	†						484	484	**484**	*	1986.7	484	-	481	481.0	-	‡	122
mgval_.30_10B	441^{E}	441^{G}		**441**	**435**	1.4	†						441	441	**441**	*	14180.09	441	543	439	439.0	21.2	‡	400
mgval_.30_10C	476^{G}	475^{G}	0.2	**478**	**464**	2.9	†						483	**476**	**475**	0.2	†	476	574	470	469.6	19.9	‡	270
mgval_.30_10D	539^{H}	531^{G}	1.5										**575**	640	530.8	17.1	†	**539**	576	528	527.0	8.7	‡	1816
Number of new best known values				25		9							9	1		10		8	3	4		0		

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TABLE 12. Computational results for Mgval {0.35}

	Best known			[15]				[3]					[16]	[30]				[21]	BLW (This paper)					
Instance	UB	LB	Gap	UB	Root	Gap'	Time	UB	LB	Root	Gap	Time	UB	UB	Root	Gap'	Time	UB	UB	LB	Root	Gap	Time	Nodes
mgval_.35_1A	158^{E}	158^{E}		**158**	**158**	*	0.2						158	158	158	*	7.5	158	158	158	158.0	*	7130.1	2000
mgval_.35_1B	192^{E}	192^{E}		**192**	188	*	39.6						192	192	190	*	313.0	192	192	192	190.0	*	260.2	298
mgval_.35_1C	283^{J}	276^{J}	2.5										**284**	312	**272**	12.8	†	289	**283**	**276**	267.9	2.5	‡	58324
mgval_.35_2A	286^{E}	286^{E}		**286**	280.5	*	2.0						286	286	286	*	9.5	286	286	285	277.4	0.4	3232.8	1088
mgval_.35_2B	326^{E}	326^{E}		**326**	305.67	*	135.1						326	326	326	*	577.0	326	332	314	313.6	5.6	‡	7300
mgval_.35_2C	485^{H}	482^{G}	0.6										**523**	520	**482**	7.3	†	**485**	485	480	470.0	1.0	‡	92732
mgval_.35_3A	84^{E}	84^{E}		**84**	81.67	*	2.9						84	84	84	*	14.7	84	84	83	81.6	1.2	‡	2278
mgval_.35_3B	113^{E}	113^{E}		**113**	105	*	129.0						113	113	113	*	33.0	113	113	108	107.5	4.5	‡	23106
mgval_.35_3C	150^{E}	149^{G}	0.7	**150**	**130**	6.8	†						159	150	**149**	0.7	†	150	150	149	148.0	0.7	‡	94190
mgval_.35_4A	430^{E}	430^{G}		**430**	**415**	1.9	†						430	430	**430**	*	449.4	430	516	424	423.5	19.6	‡	320
mgval_.35_4B	531^{E}	529^{G}	0.4	**531**	**488**	5.9	†						531	531	**529**	0.4	†	531	630	518	517.5	19.5	‡	1280
mgval_.35_4C	516^{E}	516^{G}		**516**	**479.2**	7.1	†						553	516	**516**	*	13985.6	516	531	507	506.3	4.6	‡	1128
mgval_.35_4D	643^{H}	641^{G}	0.3										**661**	729	640.5	12.1	†	**643**	647	636	635.1	1.7	‡	4946
mgval_.35_5A	454^{E}	454^{G}		**454**	**436**	3.3	†						454	454	**454**	*	2396.5	455	510	452	447.6	12.1	‡	280
mgval_.35_5B	467^{E}	466^{G}	0.2	**467**	**439**	4.6	†						468	467	**466**	0.2	†	467	486	454	453.3	6.8	‡	588
mgval_.35_5C	586^{E}	579^{G}	1.2	**586**	**538.5**	7.8	†						595	586	**579**	1.2	†	586	587	577	576.7	1.7	‡	1390
mgval_.35_5D	588^{H}	568^{G}	3.5										**648**	658	**568**	13.7	†	**588**	613	564	561.1	8.3	‡	5698
mgval_.35_6A	248^{E}	248^{E}		**248**	246	*	50.2						248	248	248	*	126.4	248	248	246	245.0	0.8	2889.8	594
mgval_.35_6B	250^{E}	250^{E}		**250**	241	*	1715.7						250	250	250	*	717.6	250	252	245	242.6	2.8	‡	4218
mgval_.35_6C	312^{H}	307^{J}	1.6										**326**	372	**303**	18.6	†	**312**	312	**307**	302.9	1.6	‡	58500
mgval_.35_7A	264^{E}	264^{E}		**264**	**264**	*	1.2						264	264	264	*	134.3	264	273	263	262.0	3.7	‡	1242
mgval_.35_7B	325^{E}	325^{E}		**325**	321	*	208.8						325	325	325	*	1756.4	325	376	325	324.6	14.6	‡	2074
mgval_.35_7C	336^{H}	336^{G}											**351**	390	**336**	13.9	†	**336**	338	331	330.2	2.1	‡	15762
mgval_.35_8A	415^{E}	415^{G}		**415**	**414**	0.2	†						415	415	**415**	*	296.0	415	415	409	409.0	1.5	‡	642
mgval_.35_8B	385^{E}	385^{G}		**385**	**376**	2.3	†						385	385	**385**	*	999.0	385	429	380	378.8	12.1	‡	770
mgval_.35_8C	489^{J}	489^{J}											**547**	602	**487**	19.1	†	**499**	**489**	**489**	487.0	*	935.4	1134
mgval_.35_9A	324^{E}	324^{E}		**324**	322	*	1071.3						324	324	324	*	2698.2	324	-	320	320.0	-	‡	386
mgval_.35_9B	331^{G}	331^{G}		**332**	**320.5**	3.5	†						332	**331**	**331**	*	12717.9	331	366	324	324.0	12.2	‡	590
mgval_.35_9C	328^{H}	328^{G}		**329**	**316**	4.0	†						338	329	**328**	0.3	†	**328**	-	323	322.5	-	‡	492
mgval_.35_9D	430^{H}	422^{G}	1.9										**473**	491	**422**	14.1	†	**430**	463	418	417.2	10.2	‡	5408
mgval_.35_10A	475^{E}	475^{E}		**475**	472	*	2230.3						475	475	475	*	7015.3	479	560	474	473.1	16.6	‡	196
mgval_.35_10B	461^{E}	461^{G}		**461**	**457**	0.9	†						463	461	**461**	*	9490.7	461	546	458	457.6	17.5	‡	488
mgval_.35_10C	430^{H}	428^{G}	0.5	**431**	**411**	4.6	†						448	431	**428**	0.7	†	**430**	-	419	418.4	-	‡	268
mgval_.35_10D	524^{H}	519^{G}	1.0										**566**	594	**519**	12.6	†	**524**	533	504	502.8	5.6	‡	1670
Number of new best known values				25		12							9	1		7		10	2	3		1		

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TABLE 13. Computational results for Mgval {0.40}

	Best known			[15]				[3]					[16]	[30]				[21]	BLW (This paper)					
Instance	UB	LB	Gap	UB	Root	Gap'	Time	UB	LB	Root	Gap	Time	UB	UB	Root	Gap'	Time	UB	UB	LB	Root	Gap	Time	Nodes
mgval_.40_1A	165^{E}	165^{E}		**165**	164	*	1.3						165					165	186	164	164.0	12.6	‡	3608
mgval_.40_1B	196^{E}	196^{E}		**196**	190	*	1415.0						196					196	196	186	186.0	5.2	‡	8408
mgval_.40_1C	263^{H}	259^{J}	1.5										**272**					**263**	263	**259**	250.7	1.5	‡	55036
mgval_.40_2A	222^{E}	222^{E}		**222**	212	*	1.7						222					222	222	221	221.0	0.5	1507.2	636
mgval_.40_2B	311^{E}	311^{E}		**311**	276	*	1506.0						311					311	311	305	304.8	1.9	‡	3488
mgval_.40_2C	469^{H}	469^{J}											**485**					**469**	469	**469**	463.2	*	3348.9	15134
mgval_.40_3A	86^{E}	86^{E}		**86**	**86**	*	0.4						86					86	86	86	85.0	*	164.6	160
mgval_.40_3B	110^{E}	110^{E}		**110**	107	*	7.3						110					110	110	107	106.8	2.8	‡	13342
mgval_.40_3C	148^{E}	148^{J}		**148**	**120**	12.8	†						157					148	148	**148**	144.1	*	2464.2	26444
mgval_.40_4A	400^{E}	400^{E}		**400**	384	*	11152.6						400					400	456	394	393.9	14.6	‡	350
mgval_.40_4B	423^{E}	418^{J}	1.2	**423**	**395**	4.3	†						423					423	-	**418**	417.5	-	‡	356
mgval_.40_4C	462^{E}	452^{J}	2.2	**462**	**424**	6.9	†						487					462	472	**452**	451.0	4.3	‡	846
mgval_.40_4D	624^{H}	612^{J}	1.9										**669**					**624**	634	**612**	609.0	3.5	‡	4320
mgval_.40_5A	426^{E}	426^{E}		**426**	423.33	*	195.3						426					426	-	425	424.5	-	‡	390
mgval_.40_5B	424^{E}	413^{J}	2.6	**424**	**401.78**	4.2	†						428					424	434	**413**	411.6	5.0	‡	852
mgval_.40_5C	524^{E}	522^{J}	0.4	**524**	**487.83**	6.9	†						539					527	598	**522**	517.5	13.6	‡	1536
mgval_.40_5D	614^{H}	595^{J}	3.1										**665**					**614**	652	**595**	593.0	9.1	‡	6394
mgval_.40_6A	224^{E}	224^{E}		**224**	220.5	*	150.2						224					224	235	222	221.1	5.7	‡	2340
mgval_.40_6B	211^{E}	211^{E}		**211**	203.22	*	858.3						211					211	211	207	206.2	1.9	‡	3348
mgval_.40_6C	312^{H}	303^{J}	2.9										**316**					**312**	312	**303**	300.9	2.9	‡	42814
mgval_.40_7A	271^{E}	271^{E}		**271**	269	*	956.9						271					271	-	265	265.0	-	‡	1514
mgval_.40_7B	270^{E}	270^{E}		**270**	258.68	*	1609.0						270					270	275	264	263.8	4.1	‡	1076
mgval_.40_7C	332^{H}	320^{J}	3.7										**336**					**332**	335	**320**	318.3	4.6	‡	11042
mgval_.40_8A	393^{E}	393^{E}		**393**	391	*	4331.6						393					393	471	393	393.0	18.1	‡	472
mgval_.40_8B	371^{E}	367^{J}	1.1	**371**	**356**	3.4	†						372					372	469	**367**	366.1	24.4	‡	550
mgval_.40_8C	517^{H}	511^{J}	1.2										**573**					**517**	517	**511**	509.4	1.2	‡	8662
mgval_.40_9A	341^{E}	337^{E}	1.2	**341**	**337**	0.7	†						341					341	-	336	335.5	-	‡	214
mgval_.40_9B	327^{E}	325^{J}	0.6	**327**	**319**	2.5	†						331					327	360	**325**	324.5	10.2	‡	478
mgval_.40_9C	295^{E}	290^{J}	1.7	**295**	**280**	5.1	†						301					295	-	**290**	290.0	-	‡	588
mgval_.40_9D	383^{H}	372^{J}	2.9										**414**					**383**	427	**372**	371.5	13.8	‡	1746
mgval_.40_10A	406^{E}	406^{E}		**406**	404	*	5107.9						406					406	-	406	405.2	-	‡	44
mgval_.40_10B	433^{E}	430.5^{E}	0.6	**433**	**430.5**	0.6	†						439					433	-	430	429.7	-	‡	224
mgval_.40_10C	432^{E}	427^{J}	1.2	**432**	**417**	2.8	†						435					434	-	**427**	426.8	-	‡	188
mgval_.40_10D	484^{H}	475^{J}	1.9										**521**					**484**	496	**475**	474.1	4.3	‡	772
Number of new best known values				25		14							9					9	0	18		2		

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TABLE 14. Computational results for Mgval {0.45}

	Best known			[15]				[3]					[16]	[30]				[21]	BLW (This paper)					
Instance	UB	LB	Gap	UB	Root	Gap'	Time	UB	LB	Root	Gap	Time	UB	UB	Root	Gap'	Time	UB	UB	LB	Root	Gap	Time	Nodes
mgval_.45_1A	168^E	168^E		**168**	166	*	7.1						168					168	170	162	161.0	4.8	‡	2284
mgval_.45_1B	166^E	166^E		**166**	**166**	*	3.2						166					166	166	166	166.0	*	166.5	256
mgval_.45_1C	258^H	256^J	0.8										**313**					**258**	258	**256**	250.2	0.8	‡	70094
mgval_.45_2A	251^E	251^E		**251**	**251**	*	0.5						251					251	251	251	251.0	*	284.8	202
mgval_.45_2B	314^E	314^E		**314**	281	*	1312.0						314					314	317	304	299.7	4.2	‡	2646
mgval_.45_2C	462^H	462^J											**496**					**462**	462	**462**	452.6	*	5074.4	25242
mgval_.45_3A	82^E	82^E		**82**	78	*	4.8						82					82	82	80	79.7	2.5	‡	1800
mgval_.45_3B	91^E	91^E		**91**	87	*	3.2						91					91	91	90	88.9	1.1	‡	7576
mgval_.45_3C	143^E	143^J		**143**	**122**	8.7	†						143					143	143	**143**	140.6	*	181.5	2218
mgval_.45_4A	381^E	381^E		**381**	365	*	1198.9						381					381	470	374	373.9	22.7	‡	176
mgval_.45_4B	471^E	462^J	1.9	**471**	**423**	5.1	†						481					471	521	**462**	461.7	12.0	‡	566
mgval_.45_4C	481^E	468^J	2.7	**481**	**434**	9.2	†						508					481	539	**468**	466.8	14.1	‡	1038
mgval_.45_4D	577^H	561^J	2.8										**626**					**577**	577	**561**	560.3	2.8	‡	4672
mgval_.45_5A	391^E	381^J	2.6	**391**	**377.38**	2.9	†						391					392	460	**381**	376.6	18.8	‡	418
mgval_.45_5B	416^E	398^J	4.4	**416**	**378.61**	5.6	†						416					416	429	**398**	397.2	7.5	‡	538
mgval_.45_5C	492^E	485^J	1.4	**492**	**445**	7.8	†						502					494	503	**485**	484.8	3.6	‡	862
mgval_.45_5D	552^H	546^J	1.1										**612**					**552**	552	**546**	544.1	1.1	‡	4990
mgval_.45_6A	213^E	213^E		**213**	207	*	528.5						213					213	280	205	204.1	30.9	‡	2238
mgval_.45_6B	210^E	210^E		**210**	192.05	*	892.7						210					210	210	207	206.6	1.4	‡	5184
mgval_.45_6C	296^H	295^J	0.3										**308**					**296**	296	**295**	292.8	0.3	‡	30680
mgval_.45_7A	261^E	261^E		**261**	257.5	*	200.1						261					261	-	258	256.6	-	‡	780
mgval_.45_7B	294^E	290^E	1.4	**294**	**290**	1.3	†						294					294	-	289	288.2	-	‡	1034
mgval_.45_7C	337^H	327^J	3.0										**347**					**337**	340	**327**	325.8	3.9	‡	15596
mgval_.45_8A	370^E	369^J	0.3	**370**	**367**	0.6	†						370					370	401	**369**	367.0	8.3	‡	488
mgval_.45_8B	360^E	359^J	0.3	**360**	**341**	4.7	†						376					364	368	**359**	358.4	2.5	‡	458
mgval_.45_8C	501^J	485^J	3.2										**535**					**504**	**501**	**485**	482.1	3.2	‡	24202
mgval_.45_9A	306^E	300^J	2.0	**306**	**299**	2.1	†						306					306	-	**300**	299.7	-	‡	152
mgval_.45_9B	323^E	319^J	1.2	**323**	**311**	3.4	†						323					323	-	**319**	316.4	-	‡	424
mgval_.45_9C	291^E	289^J	0.7	**291**	**273**	5.9	†						306					291	-	**289**	288.5	-	‡	608
mgval_.45_9D	387^H	381^J	1.6										**403**					**387**	403	**381**	379.8	5.6	‡	3444
mgval_.45_10A	388^E	386^J	0.5	**388**	**385**	0.7	†						388					388	-	**386**	385.2	-	‡	94
mgval_.45_10B	399^E	396^J	0.8	**399**	**390**	2.3	†						399					399	-	**396**	395.8	-	‡	64
mgval_.45_10C	403^E	398^J	1.2	**403**	**382**	3.8	†						418					403	-	**398**	397.8	-	‡	186
mgval_.45_10D	488^H	474^J	2.9										**504**					**488**	502	**474**	473.4	5.7	‡	2478
Number of new best known values				25		10							9					9	1	23		2		

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TABLE 15. Computational results for Mgval {0.50}

	Best known			[15]				[3]					[16]	[30]				[21]	BLW (This paper)						
Instance	UB	LB	Gap	UB	Root	Gap'	Time	UB	LB	Root	Gap	Time	UB	UB	Root	Gap'	Time	UB	UB	LB	Root	Gap	Time	Nodes
mgval_.50_1A	145^{E}	145^{E}		**145**	137	*	6.3	-	138	138.0	-	‡	145					145	166	138	138.0	18.4	‡	1152
mgval_.50_1B	170^{E}	170^{E}		**170**	**170**	*	2.9	170	170	170.0	*	459.5	170					170	170	170	170.0	*	26.8	58
mgval_.50_1C	261^{K}	254^{J}	2.7					**270**	**253**	249.0	6.5	‡	**261**					270	265	**254**	249.9	4.2	‡	74924
mgval_.50_2A	248^{E}	248^{E}		**248**	**248**	*	0.6	351	234	233.5	40.0	‡	248					248	248	248	240.0	*	292.2	238
mgval_.50_2B	284^{E}	284^{E}		**284**	262	*	279.0	-	274	273.8	-	‡	284					284	284	283	282.2	0.4	‡	4660
mgval_.50_2C	464^{H}	459^{J}	1.1					473	**453**	447.3	4.3	‡	488					**464**	464	**459**	448.6	1.1	‡	90996
mgval_.50_3A	75^{E}	75^{E}		**75**	71.09	*	1.6	-	73	72.1	-	‡	75					75	75	74	72.8	1.3	‡	2734
mgval_.50_3B	107^{E}	107^{E}		**107**	98.5	*	801.4	-	101	100.3	-	‡	107					107	107	101	100.3	5.8	‡	18408
mgval_.50_3C	137^{E}	137^{I}		**137**	**116.25**	7.9	†	137	**137**	135.1	*	9.8	139					137	137	136	135.4	0.7	4.6	120
mgval_.50_4A	350^{E}	350^{E}		**350**	346	*	144.9	-	346	346.0	-	‡	350					350	438	348	347.3	22.9	‡	510
mgval_.50_4B	413^{E}	400^{I}	3.2	**413**	**361**	4.2	†	-	**400**	398.1	-	‡	419					414	491	400	399.7	20.4	‡	624
mgval_.50_4C	488^{E}	479^{J}	1.9	**488**	**441.5**	9.1	†	-	**472**	471.2	-	‡	512					488	502	**479**	478.4	4.7	‡	1152
mgval_.50_4D	587^{H}	568^{J}	3.3					-	**565**	563.6	-	‡	**613**					**587**	608	**568**	566.0	6.8	‡	5804
mgval_.50_5A	367^{E}	367^{E}		**367**	360	*	1296.0	-	360	359.4	-	‡	367					367	381	361	360.2	5.4	‡	290
mgval_.50_5B	377^{J}	370^{J}	1.9	**378**	**348**	3.7	†	-	**365**	364.6	-	‡	378					378	**377**	**370**	365.4	1.9	‡	420
mgval_.50_5C	457^{H}	455^{J}	0.4	**459**	**416.5**	8.4	†	-	**449**	448.6	-	‡	476					**457**	459	**455**	454.4	0.9	‡	1112
mgval_.50_5D	541^{J}	531^{J}	1.9					551	**527**	525.3	4.5	‡	570					**544**	**541**	**531**	528.7	1.9	‡	5382
mgval_.50_6A	210^{E}	210^{E}		**210**	199	*	123.2	-	198	197.3	-	‡	210					210	229	201	200.2	13.0	‡	4418
mgval_.50_6B	210^{E}	210^{E}		**210**	192.5	*	368.1	-	206	205.4	-	‡	210					210	210	207	205.9	1.4	‡	7734
mgval_.50_6C	293^{H}	289^{J}	1.4					-	**281**	279.7	-	‡	**306**					**293**	293	**289**	284.1	1.4	‡	57698
mgval_.50_7A	248^{E}	248^{E}		**248**	243	*	126.8	-	237	236.2	-	‡	248					248	-	243	241.2	-	‡	862
mgval_.50_7B	276^{E}	276^{E}		**276**	272	*	1429.5	-	271	268.9	-	‡	276					276	283	271	271.0	4.3	‡	2552
mgval_.50_7C	320^{H}	308^{J}	3.8					-	**306**	304.6	-	‡	**333**					**320**	321	**308**	304.7	4.1	‡	10960
mgval_.50_8A	388^{E}	387^{J}	0.3	**388**	**382**	1.4	†	-	**386**	385.6	-	‡	388					388	390	**387**	386.9	0.8	‡	278
mgval_.50_8B	350^{E}	345^{J}	1.4	**350**	**330**	4.7	†	-	**343**	342.5	-	‡	356					350	363	**345**	343.5	5.1	‡	700
mgval_.50_8C	501^{H}	486^{J}	3.0					502	**485**	483.9	3.4	‡	535					**501**	502	**486**	483.9	3.2	‡	17290
mgval_.50_9A	306^{E}	306^{E}		**306**	**306**	*	4.1	-	305	304.1	-	‡	306					306	-	306	306.0	-	‡	114
mgval_.50_9B	278^{E}	268^{J}	3.7	**278**	**261.65**	3.5	†	-	**267**	266.7	-	‡	278					278	347	**268**	267.9	25.7	‡	566
mgval_.50_9C	293^{H}	286^{J}	2.4	**301**	**278.33**	7.4	†	-	**283**	281.9	-	‡	303					**293**	-	**286**	285.7	-	‡	602
mgval_.50_9D	361^{H}	353^{J}	2.2					-	**349**	348.9	-	‡	**408**					**361**	380	**353**	351.7	7.4	‡	2184
mgval_.50_10A	385^{E}	381^{J}	1.0	**385**	**377.5**	1.3	†	-	-	-	-	‡	385					385	485	**381**	380.8	24.0	‡	100
mgval_.50_10B	369^{E}	367^{J}	0.5	**369**	**364**	1.4	†	-	-	-	-	‡	371					369	-	**367**	366.4	-	‡	156
mgval_.50_10C	406^{E}	397^{J}	2.2	**406**	**389**	4.2	†	-	**396**	395.7	-	‡	416					406	-	**397**	396.9	-	‡	516
mgval_.50_10D	456^{H}	440^{J}	3.6					-	**436**	433.3	-	‡	**492**					**456**	499	**440**	436.1	12.6	‡	908
Number of new best known values				25		13		1	19		1		6					10	2	19		0		

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